

CHAPTER 11

ANIMAL SYSTEMS

FOR LIFE



By the end of this chapter you will have covered the following material.

Science Understanding

- In animals, the exchange of gases between the internal and external environments of the organism is facilitated by the structure and function of the respiratory system at cell and tissue levels (ACSBL056)
- In animals, the exchange of nutrients and wastes between the internal and external environments of the organism is facilitated by the structure and function of the cells and tissues of the digestive system (for example, villi structure and function), and the excretory system (for example, nephron structure and function) (ACSBL057)
- In animals, the transport of materials within the internal environment for exchange with cells is facilitated by the structure and function of the circulatory system at cell and tissue levels (for example, the structure and function of capillaries) (ACSBL058)





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Figure 11.1 ►

A Southern right whale showing baleen, which is used to separate krill and plankton from sea water

All organisms have adaptations for survival. Many of these are obviously related to improving their chances of getting food. Whales of the suborder Mysticeti do not have teeth, but possess baleen plates instead. These keratinised bristly plates can be 0.5–3.5 m long and there can be up to 400 in a row, creating a stiff yet flexible sieve-like structure to separate their tiny prey of krill and plankton (copepods). From there, these small organisms need to be broken down into molecules small enough to get into cells in order for the whale to survive. A series of systems, organs, tissues and structures function together to move the carbon and nitrogen atoms of each copepod into living cells.

Transporting substances: the human circulatory system

The role of hormones is discussed in further detail in Unit 4 Biology.

The **circulatory system** is the transport system of the human body and has four principal functions:

- 1 transportation of water, oxygen and carbon dioxide
- 2 distribution of nutrients and removal of wastes
- 3 maintenance of body temperature
- 4 circulation of hormones.

The essential features of a transport system must include the ability to reach every cell in the body, distributing the necessary requirements in a usable form and not disrupting the functioning of other systems in the body. The circulatory system consists of three general components that incorporate each of these essential features:

- 1 a fluid in which materials are transported, such as blood
- 2 a system of blood vessels or spaces throughout the body in which the fluid moves
- 3 a pump, such as the heart, that pushes the fluid through the blood vessels or spaces.

The heart, blood and blood vessels make up the **cardiovascular system**.



QUIZ

Take the quiz to see how much you already know about the cardiovascular system.

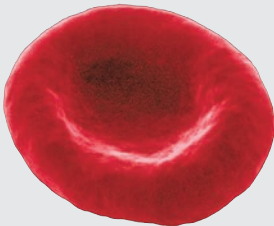
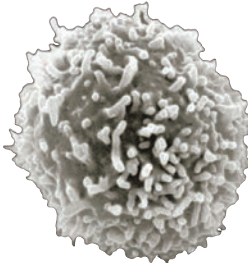
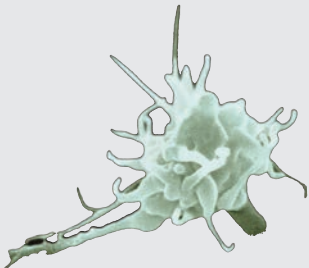
Circulatory fluid: the blood

The average 70 kg adult human is nourished and protected by about 5 L of blood.

Approximately 55% of the blood is liquid, called the **plasma**; the remaining 45% is composed of blood cells, including **erythrocytes**, **leukocytes** and **platelets**. The blood cells are considered to be a tissue because they are cells that work together for a common cause.

Table 11.1 summarises the components of blood.

Table 11.1 Structure and function of the components of the blood tissue

Blood component	Structure and function
Plasma  Getty Images/SPL	Pale yellow fluid component of the blood through which the solid blood components, along with body heat, dissolved gases, nutrients, wastes and hormones, are transported around the body Contains a number of dissolved proteins that have different functions. These plasma proteins carry other molecules, such as calcium, drugs, hormones, lipids, vitamins and cholesterol. They also help keep the acidity of the blood at the optimum level (between pH 7.35 and 7.45). One protein is essential in blood clotting. Another group contains antibodies that provide immunity against diseases such as measles. Others regulate the balance of fluids in the plasma.
Erythrocytes (red blood cells)  Corbis/Visuals Unlimited/Dr Stanley Flegler	Contain haemoglobin, a molecule that enables the cell to bind oxygen molecules Without haemoglobin, red blood cells would supply enough oxygen to support life for about 4.5 seconds. With haemoglobin, life can continue for about 5 minutes. (This seems like a short time, but blood is continuously transported back to the heart and then to the lungs, where the oxygen supply is renewed.) Do not have a nucleus in their mature stage, allowing the cell to carry more haemoglobin
Leukocytes (white blood cells)  Corbis/Visuals Unlimited/Dr Stanley Flegler	Function mainly to protect the body against invading micro-organisms and toxins Are much fewer in number than red blood cells Are of several types and, unlike red blood cells, contain a nucleus Some types destroy micro-organisms by engulfing them, and then using enzymes to digest the micro-organism and the leukocyte itself; others are specialised in producing antibodies that are an important part of the body's immune system. Fragments of white blood cells and invading micro-organisms are called pus
Platelets  Corbis/Visuals Unlimited/Dr Stanley Flegler	Small fragments of larger cells in bone marrow Function primarily to initiate blood clotting Are irregularly shaped and move easily through smooth blood vessels when resting Become active and rupture if they encounter a sharp edge, such as a cut, as they move through a blood vessel When the platelet membrane breaks, the platelet releases a substance that reacts with proteins in the plasma to create a mesh of fibres This mesh prevents further blood flow. After a few days, fibres contract and begin to close the wound to the blood vessel.

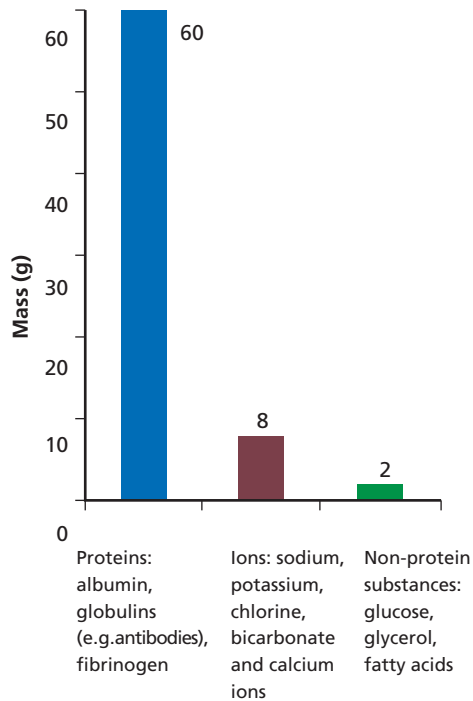


Figure 11.2 ▲
Graph comparing the main substances (apart from water) in 1 L of plasma

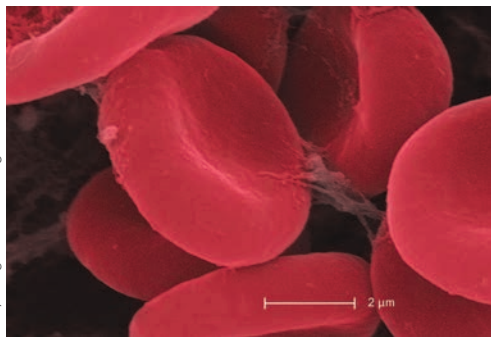


Figure 11.3 ▲
A scanning electron micrograph of red blood cells (and fibrin, the protein formed during the clotting of blood)

Chapter 7 discusses the process and products of cellular respiration.

WOW

The colour of blood

Nearly all vertebrates use haemoglobin in blood cells to transport oxygen around the body, hence the blood is bright red when oxygenated and a deep burgundy colour when deoxygenated. There is a group of fish, however, that have colourless blood. The Antarctic ice fish lack haemoglobin and thus their blood is transparent. One explanation for this relates to their environment: the ocean water they live in varies from 2°C to -2°C. However, it does not freeze due to the solutes in the water that reduce its freezing point. The blood contains a type of anti-freeze made of glycoproteins. It is these that give the blood its transparency. Insects also have colourless blood, and marine worms have green blood.

Red blood cells and oxygen transport

There are about 5 million red blood cells (or erythrocytes) per cubic millimetre of blood in a healthy human adult. Each cell is shaped like a disc with both flat sides pressed in, and measures about 8 μm in diameter and about 2 μm thick.

Red blood cells are produced in the bone marrow. Just before they are released from the bone marrow and move into the circulatory system, their nucleus breaks down. This makes them pliable and elastic so that they can twist and flex as they move through blood vessels that are sometimes narrower than their unbent size. The lack of nuclei also allows more space to carry oxygen. Red blood cells have a life span of about 4 months. The spleen and the liver break down and remove old and damaged red blood cells.

The main function of red blood cells is to carry oxygen and, to a lesser extent, carbon dioxide. Normally only about 0.3 mL of oxygen can dissolve in 100 mL of blood. However, red blood cells contain an iron-containing pigment called haem and a special protein called globin, which gives them their red colour. **Haemoglobin** enables blood to carry about 20 mL of oxygen per 100 mL of blood. This is about the same as the proportion for oxygen in air!

When there is a high concentration of oxygen, as in the capillaries exiting the lungs, the oxygen combines with the haemoglobin to form **oxyhaemoglobin**. Where there is a low concentration of oxygen, as in the tissues, the oxygen molecules **dissociate** from the haemoglobin. The oxygen diffuses through the red blood cell plasma membrane, travels through the plasma and out through the cell membranes of the capillary wall to cells with a low concentration of oxygen.

When the body is at rest, the cells use only about one-quarter of the total oxygen carried in the blood. However, with heavy exercise, more than three-quarters of the available oxygen is used. Thus, the body has a large capacity to provide extra oxygen to cells in case of emergencies.

Blood and carbon dioxide transport

Cellular respiration in each cell produces carbon dioxide as a waste product. Its removal ensures the cell pH level is maintained for optimal function. Just as the bloodstream is able to deliver oxygen to every cell in the body, it is the bloodstream that removes the carbon dioxide from every cell in the body.

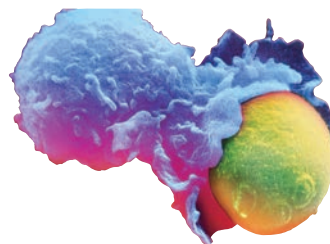
Carbon dioxide is quite soluble and reacts with water to form **carbonic acid**. This is normally a slow reaction. In the red blood cell, however, where the initial reaction occurs, the enzyme carbonic anhydrase speeds up the reaction. Approximately 70% of the carbon dioxide produced by cells is carried as carbonic acid in the plasma back to the lungs. The rest is either carried in the plasma as carbon dioxide (7%) or attaches to haemoglobin in red blood cells (23%).

As the blood passes through the lungs, the lowered concentration gradient for carbon dioxide causes all the previous reactions to reverse. The carbonic acid reverts back to carbon dioxide and water. The carbon dioxide leaves the **internal environment** of the blood, passing into the **external environment** of the lungs where it and the water are exhaled as vapour.

White blood cells

Apart from ensuring oxygen and carbon dioxide are transported effectively around the body, the bloodstream also ensures the body is protected against any sort of invasion by foreign bodies. The cells that carry out this function are the white blood cells or leukocytes. White blood cells are much larger than red blood cells and do not contain the coloured pigment haemoglobin. In order to be seen clearly under the microscope they need to be stained. They all have a nucleus, which helps in classifying them into different groups. When an infection is being fought there are about 10 000 white blood cells per cubic millimetre of blood.

Leukocytes function collectively to fight foreign substances that enter the body. Some white blood cells live from only a few minutes to days, and others can live for years. They are found in the tissues as well as the blood because they can move themselves around, like an amoeba. They can pass through capillaries by squeezing between the cells that make up the wall of the capillary and so get to regions of damaged cells.



Getty Images/Photo Researchers

Unit 4 Biology goes into more detail about the role of leukocytes in immunity.

◀ **Figure 11.4**

A scanning electron micrograph of a white blood cell engulfing a yeast cell

Platelets

In the case of an injury where bleeding is involved, platelets stem the flow of blood from the body. Platelets are smaller than red blood cells. They are fragments of cells, shaped like flattened discs, which arise in the bone marrow. They release chemicals at the site of injury in a blood vessel, causing it to contract to reduce blood loss. Platelets also clump together to help form a clot. They have a life span of one or two weeks.



BLOOD CLOTS

Read through the information and prepare a summary of how blood clots form.

QUESTION SET 11.1

Remembering

- 1 State what plasma is and list the substances carried in plasma.
- 2 Describe the roles of plasma proteins.
- 3 List the functions performed by white blood cells.
- 4 Outline the features of platelets and describe their functions.

Understanding

- 5 Explain the benefits of having haemoglobin in red blood cells.
- 6 Explain what is meant by the internal and external environment of a single cell in a multicellular organism. Name the barrier between the two environments.

Circulatory system: the blood vessels

The bloodstream in mammals is said to be a **closed circulatory system**. The blood is confined to a series of paths or vessels. The driving force behind the blood flow around the body is the heart, so all vessels must be connected to this organ. There are three main types of blood vessels. **Arteries** take blood away from the heart, **veins** carry blood to the heart and **capillaries** are found between these two blood vessels. Therefore, the major organs of the circulatory system are the heart, arteries, veins and capillaries.

Figure 11.5 shows a cross-section of the walls of an artery and a vein. It can be seen that the walls of the artery are thicker than those of the vein. This is because arteries have a lot more muscle and elastic fibres than veins. The muscle and fibre tissues of the arteries lead to the production of blood pressure as the heart beats, and forces the blood through the narrow central

cavity, the **lumen**. Arteries swell as each wave of blood is ejected from the heart. They then recoil elastically to propel the blood further on. Arteries and **arterioles** (small arteries) can also narrow when nerves in the vessel walls or chemicals carried in the blood cause the circular muscles in the walls to contract.

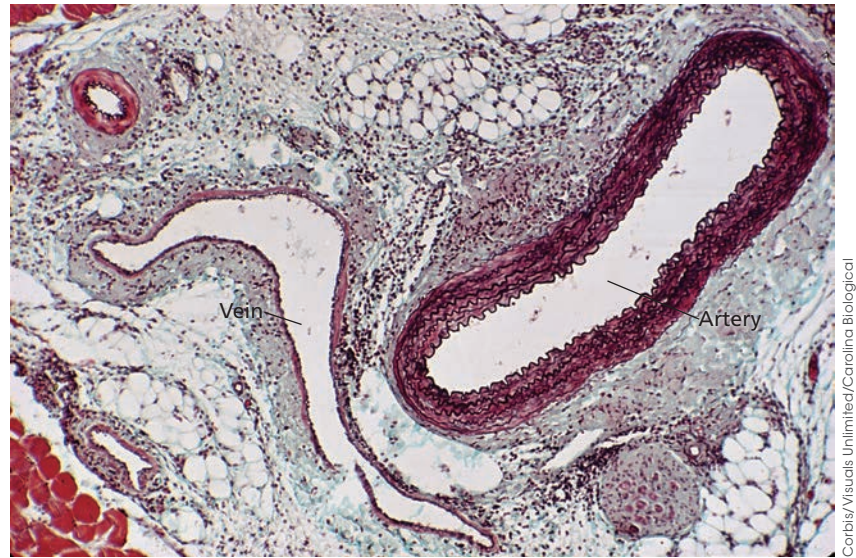


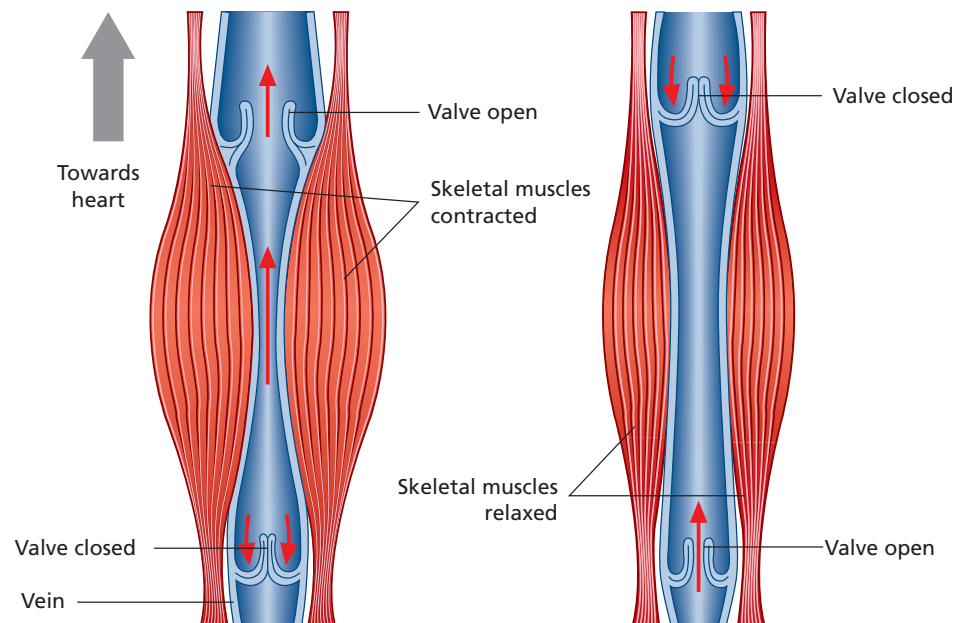
Figure 11.5 ►
Cross-section of an artery and a vein

The lumen of the vein is larger than that of the artery. More than 60% of the blood is distributed in the veins, which could be seen to act as a reservoir or as extra volume to be pumped around with increased exercise.

Veins have valves in them to help keep the blood flowing to the heart. This is especially important in the legs. While the legs are moving, the blood is squashed in the veins that travel between the muscles. The blood can only move up as the valves prevent it dropping back.

It is useful to know this as people who have to stand still for a long time in hot weather (such as soldiers on parade) sometimes feel faint, as too much blood is staying in the veins in their legs and not returning enough blood to the heart in a timely manner, thus depriving the brain of enough oxygenated blood. By continually contracting and relaxing the muscles in their legs (e.g. by wiggling their toes), they can move the blood up to their heart and avoid fainting.

Figure 11.6 ►
Contraction of skeletal muscles, in conjunction with the presence of valves in the veins, helps blood return to the heart.



Unlike arteries or veins, capillaries have very thin walls and are very narrow. In fact, their walls are only one cell thick. This makes it possible for diffusion of all the nutrients that a cell needs, such as glucose and oxygen, to occur. Wastes, such as carbon dioxide and urea, are also

able to pass from the cells to the blood. The diameter of capillaries is about the same as or smaller than a red blood cell, which further increases the diffusion rate of gases. Capillaries are so numerous and so small that every cell in the body is only a few millimetres away from a capillary.

Circulatory pump: the heart

Blood is pumped around the body by the heart. The heart is made of muscle and, in humans, it can contract continuously about 70 times a minute, day and night, for as long as we are alive. Cardiac tissue is found only in the heart. It is the coordinated contractions of cardiac muscle, or myocyte cells, that provide the pumping action of the heart.

The heart has four chambers. Two are thin-walled **atria** (singular, atrium) and the other two are the more muscular **ventricles**.

Deoxygenated blood enters the right atrium, via the **vena cava**, from the general body system of blood vessels, the **systemic circulation**, and passes into the right ventricle. When the right ventricle contracts, blood is forced into the pulmonary artery, which leads to the lungs. This is called **pulmonary circulation**. Here, oxygen diffuses into the blood and carbon dioxide passes out into the air in the lungs. The now oxygen-rich or oxygenated blood passes back into the heart through the left atrium and then into the left ventricle. When this ventricle contracts, the blood is forced into the **aorta**, the largest artery. Through its many branches, the aorta takes blood to all parts of the body. The **septum** keeps the deoxygenated blood on the right side of the heart from mixing with the oxygenated blood on the left side.

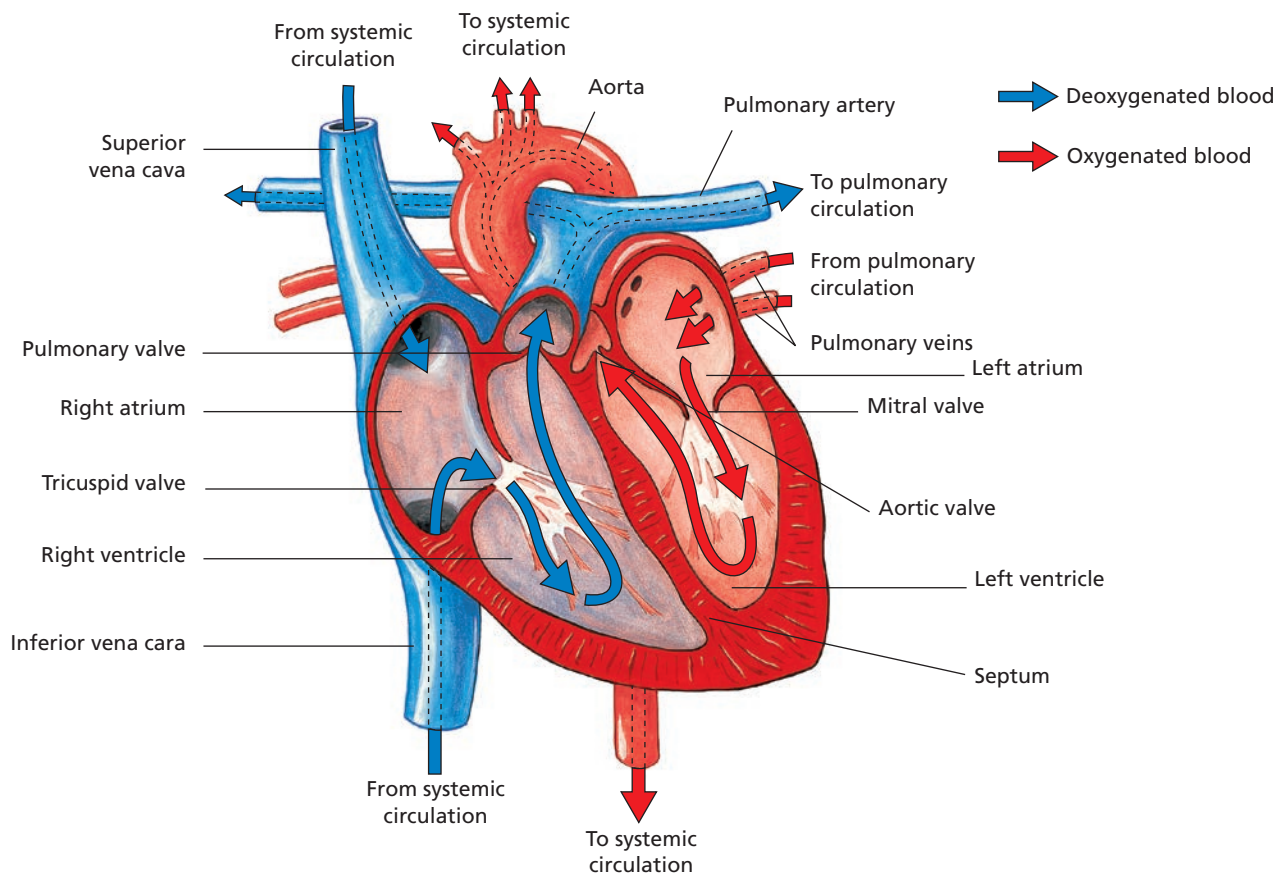
The blood is kept flowing in one direction in the heart by the presence of four **valves**: one between each atrium and ventricle (the two **atrioventricular valves**) and one between each ventricle and the artery it leads to (the **pulmonary valve** and **aortic valve**). The two ventricles contract at the same time and the atria contract at the same time, but the ventricles and atria contract at different times. This leads to the two atrioventricular valves snapping shut at the same time as the ventricles contract. The pulmonary and aortic valves close when the ventricle finishes contracting, so that blood under pressure from the enlarged elastic arteries does not flow back into the heart.



MOVING MUSCLE CELLS

View beating myocytes in this video clip.

▼ **Figure 11.7**
The human heart





SHEEP HEART DISSECTION

Complete the worksheet, either before or after your sheep heart dissection.

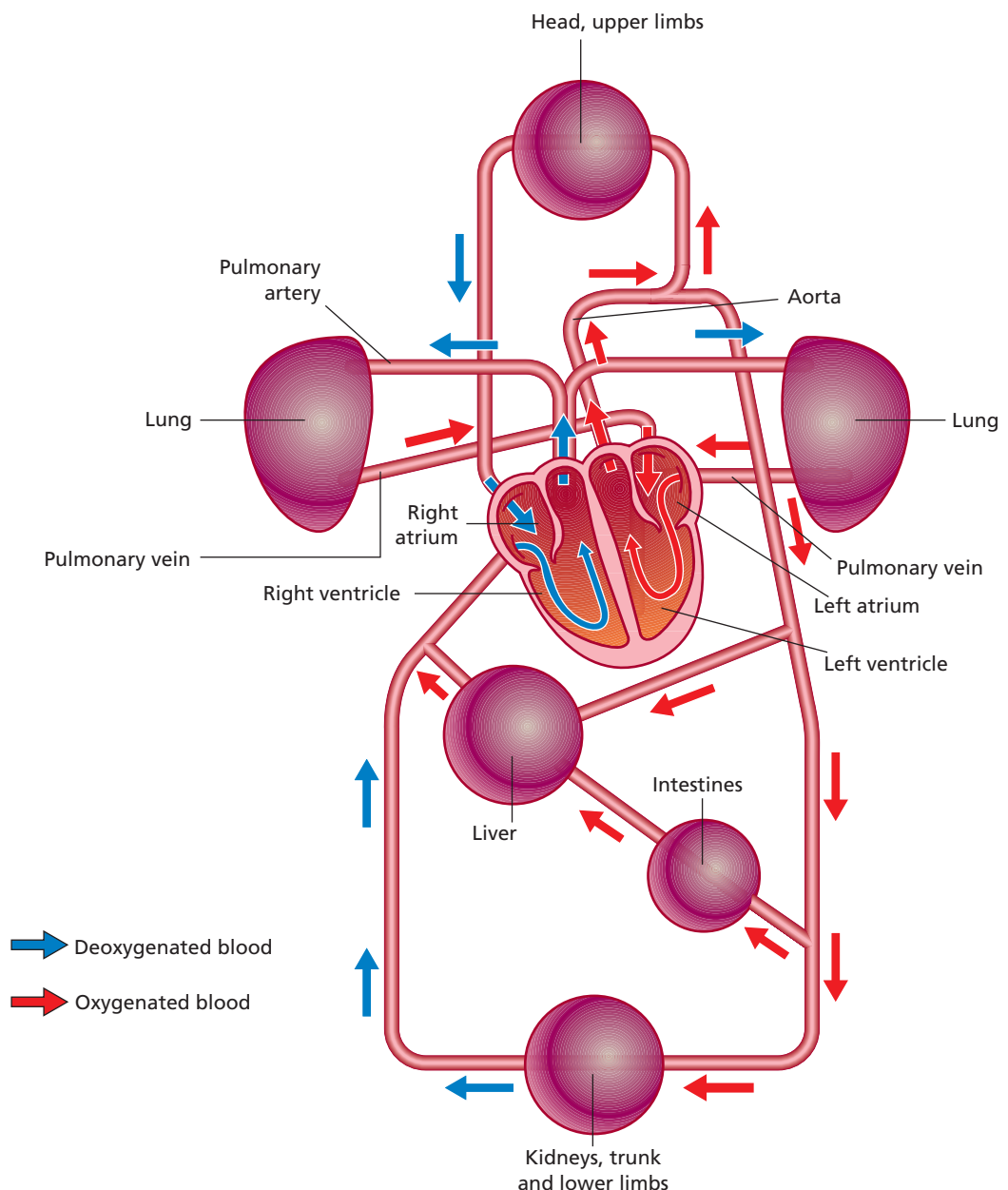
When you use a stethoscope to listen to the heart, it is the sound of the valves closing and the sudden stop of blood flow that you hear. The trained listener can determine if particular valves are not shutting completely and if the timing of valve closure is abnormal.

The amount of blood pumped through the heart in a resting 70 kg adult is about 5 L per minute, which, as mentioned earlier, is equal to the entire blood volume of the person. During heavy exercise this can increase up to seven times that amount. This is possible partly due to the heart beating faster, from a resting pulse of about 70 beats per minute to about 210 beats per minute in young people. The ventricles also force out more blood for each contraction and do it under greater pressure (Table 11.2).

Table 11.2 Changes in heart rate and blood volume pumped at rest and during heavy exercise

	Heart rate (beats per minute)	Volume of blood pumped out for each ventricular contraction (mL)	L per minute
At rest	70	70	4.9
Heavy exercise	180	160	28.8

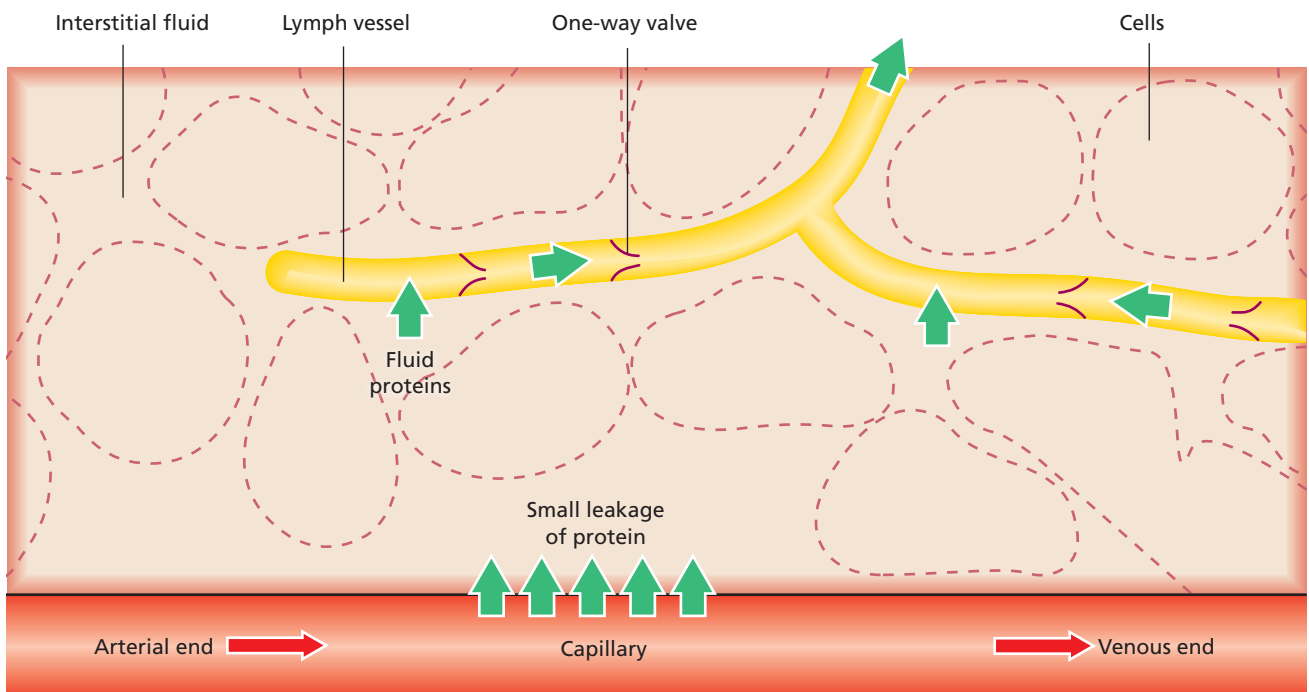
Figure 11.8 ►
Blood circulation in mammals



The lymphatic system

Gases and small molecules entering the internal environment are not the only substances to pass through capillary walls. Blood pressure forces some fluid as well as small protein molecules out of the capillaries at the arterial end. Most of the water passes back into the capillaries at the venous end, as the blood pressure has dropped off. The water moves back into the capillaries by osmosis, caused mainly by the high concentration of **albumin** molecules in the plasma.

However, some water and proteins would be left in the spaces between the cells if it were not for the **lymph vessels**. Lymphatic capillaries are small, blind-ending tubes, which allow the fluid and protein to enter through tiny flaps that act as one-way valves between the cells in their walls. The fluid flows in the lymph vessels to the heart to rejoin the normal blood circulation. It is pushed along in much the same way as blood in the veins is moved, by contracting muscles. Valves in the lymph vessel walls maintain a unidirectional flow to the heart.



People who do not get enough protein can suffer from a condition known as kwashiorkor, typically evident from severe **oedema** and swelling of tissues. This swelling up from water is caused because they do not have sufficient plasma proteins for the water to be drawn osmotically back into the capillaries.

▲ **Figure 11.9**
Fluid and proteins are removed by the lymph vessels, which, with one-way valves, transport the fluid between cells away from them.

Transport of materials in other animals

Cells with animal-like features that are smaller than 1 or 2 mm, such as an *Amoeba*, rely on diffusion across cell membranes for most of their needs. Some larger animals, such as *Planaria* and sea anemones, do not have a specialised fluid such as blood for transport, but use the fluid between the cells, the **interstitial fluid**, to carry nutrients and wastes. Other animals, such as arthropods and most molluscs, have a heart and a few vessels, but the liquid washes freely over the internal organs. This is known as an **open circulatory system**. For example, some insects have a series of little hearts that pump fluid from the rear end of the animal towards its head. The liquid leaves the open-ended aorta and flows between the cells until it is pumped forward again. As we will see, gas exchange occurs via the tracheal system in these animals, so a gas-carrying transport fluid is not necessary.

Examples of closed circulatory systems are found in all the vertebrates (including humans) as well as in squids and annelids, of which the earthworm is an example. Generally these systems are more efficient, as shown by the larger and more active animals that use them. The circulation is able to direct its flow where it is most needed; for example, to the muscles when needing to escape predators.

Fish have a two-chambered heart with one atrium and one ventricle, and oxygenated blood proceeds directly from the gills to the tissues. In mammals and birds, the oxygenated blood comes back from the lungs to the heart to be pumped out again at a much higher pressure. Amphibia have two atria, like mammals, but only one ventricle. It might seem that deoxygenated and oxygenated blood would become mixed in the single ventricle, but separation of the two bloodstreams is surprisingly complete and there is far less mixing of the blood than you would expect. Largely deoxygenated blood is sent to the lungs while mostly highly oxygenated blood is delivered to the head and brain.

The red blood cells of all vertebrates, except mammals, have nuclei. Animals vary in the type of respiratory pigment that is used. Haemoglobin is used in virtually all vertebrates. While it is not a vertebrate, the common garden earthworm is reddish because it also has haemoglobin in its blood. However, arachnids and most crustaceans have a respiratory pigment called **haemocyanin**, which contains copper instead of iron, thus making their blood appear blue when oxygenated. In many invertebrate animals, the respiratory pigment circulates freely in the plasma, whereas in all vertebrates and some invertebrates the pigment is in the cells. The advantage of having the pigment enclosed in cells is improved blood flow in small blood vessels and improved gas exchange. It also can make it possible to carry higher concentrations of substances than would be possible if they were loose in the plasma.

QUESTION SET 11.2

Remembering

- 1 Name the components and main functions of the circulatory system.
- 2 Describe features of the heart that allow it to receive blood from other parts of the body.
- 3 Define and distinguish between the circulatory system and the cardiovascular system.
- 4 Explain the events that occur in the heart to ensure that oxygenated blood is pumped around the body.

Understanding

- 5 Describe the differences between systemic and pulmonary circulation, and outline the significance of each system.
- 6 Outline the main differences between the three main types of blood vessels.
- 7 Distinguish between the lymphatic system and the circulatory system.
- 8 Explain the role of albumin in keeping fluid in circulation.
- 9 Summarise how the lymphatic system functions.

Applying

- 10 If you were shown a blood vessel attached to the heart, explain how you would know if it was an artery or vein.
- 11 Draw a simplified diagram of the heart and show, using arrows, the direction of blood flow.
- 12 Devise a table that:
 - a clearly identifies the common features of circulatory systems of different animals.
 - b compares the differences found in the circulatory systems of different animals.

Adaptations for gas exchange

All animals exchange gases with their surroundings. However, unlike water and food, gases cannot be stored easily within living tissues, and so most animals must exchange gases with the atmosphere on a continual basis.

The mechanism of gas exchange from the external environment to the internal environment, and vice versa, is similar for all living organisms. In microscopic organisms, gas exchange occurs by diffusion across their plasma membrane. In larger terrestrial animals, the gas exchange system is inside the body to prevent dehydration of the gas exchange surfaces.

The surface across which the gas exchange occurs needs to:

- be moist, as the gases dissolve in the water and diffuse from one side of the membrane to the other
- be thin and permeable, so the gas molecules can move across it easily and quickly
- have a large surface area in relation to the volume of the organism, so as to adequately provide the gaseous requirements
- have a greater concentration of required gas on one side of the membrane than the other, so that a concentration gradient is maintained.

All cells in organisms, whether unicellular or multicellular, exchange gases with their environment.

Gas exchange in mammals

Most mammals are terrestrial and obtain all their oxygen requirements from the air. They are **endothermic** and usually active. Aerobic cellular respiration taking place in their cells generates heat as a by-product, so in order to maintain a constant body temperature, their demand for oxygen is high. The heat generated ensures that the conditions are ideal for the normal functioning of the cell's other metabolic processes. These processes rely on the functioning of enzymes, among other things, that have an optimum temperature at which they operate. The mammalian lung has evolved to meet all gaseous demands.

Air passes into the lungs through the nose and mouth. It travels past the throat and **larynx** (voice box) into the **trachea**, which is strengthened with rings of cartilage to stop it collapsing. The trachea branches into two **bronchi** (singular, bronchus), which lead to the lungs. On entering the lung tissue, each bronchus continually divides into smaller tubes called **bronchioles**. Eventually, each bronchiole ends in a cluster of tiny air sacs called **alveoli** (singular, alveolus).

The alveoli endothelial tissues not only provide a moist surface across which gas exchange occurs, but also increase the surface area of the lung tissue enormously. There are about 700 million alveoli in the human lung and they provide a surface area that is approximately the size of a tennis court.

A network of capillaries occurs around each alveolus. The thin wall of the alveolus, and the thin wall of the capillary, is the boundary between the external environment in the alveolus and the internal environment of the blood in the capillaries. Exchange between the two environments occurs by diffusion. The capillaries are so narrow that, as the red blood cells are pushed through them, they are partially squashed into a conical shape. This not only increases the amount of surface area in contact with the capillary wall, but ensures there is little fluid between the blood cell and the alveolar wall, so the distance the oxygen and carbon dioxide molecules have to travel is very short, less than 1 μm .

Each alveolus is covered, internally, with a thin layer of fluid in which the oxygen dissolves before it diffuses into the cells. This fluid contains a **surfactant**, a detergent-like **lipoprotein**, which reduces its **surface tension**. This prevents the alveolar walls from being pulled inwards so that the alveoli don't collapse and the lungs can expand.

See Chapter 9 for more on how temperature can affect the functioning of enzymes.



CIRCULATORY AND RESPIRATORY SYSTEMS

Watch the video and summarise the connection between the circulatory and respiratory systems.

Figure 11.10 ►
The gas exchange system in humans

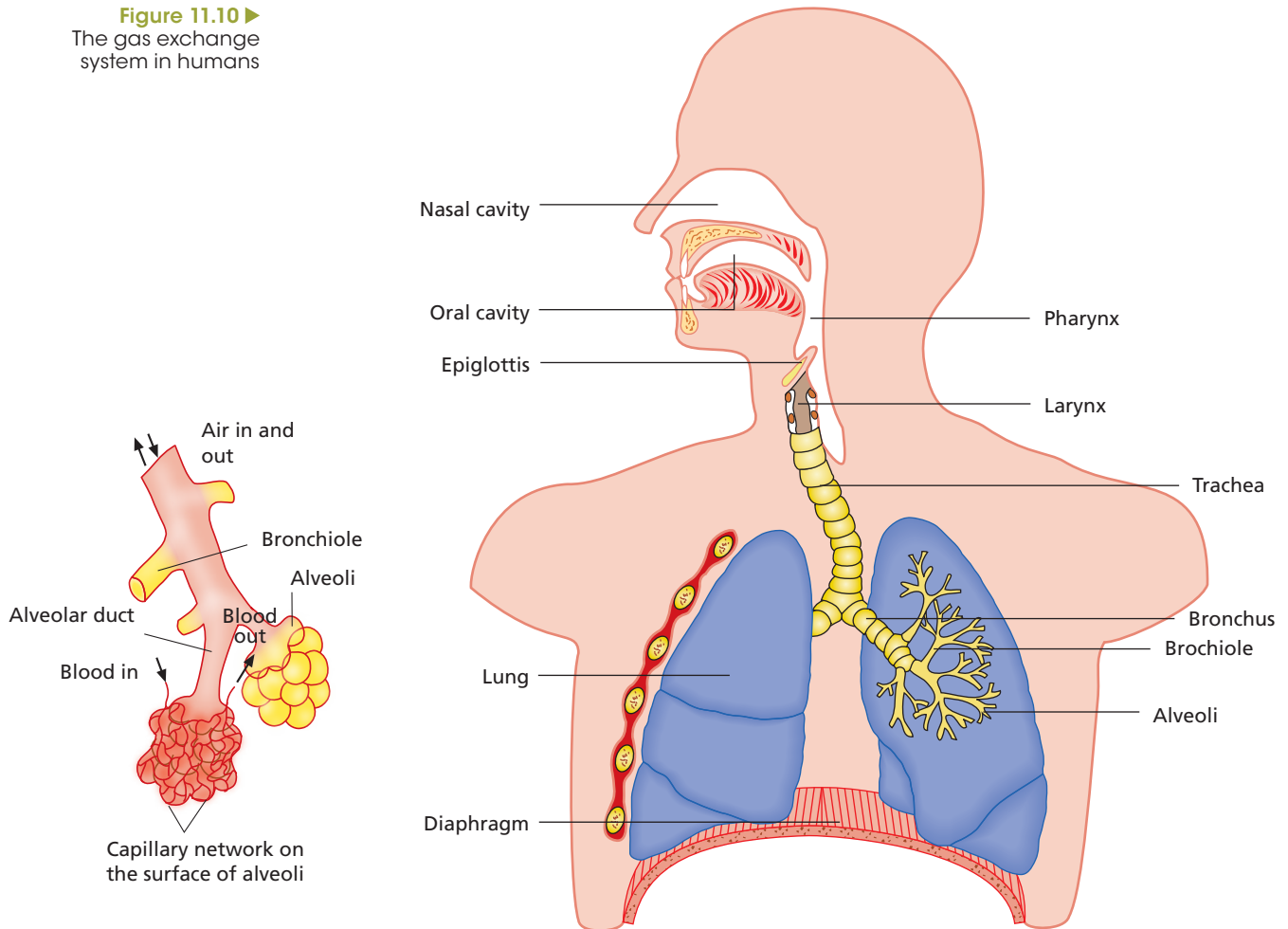
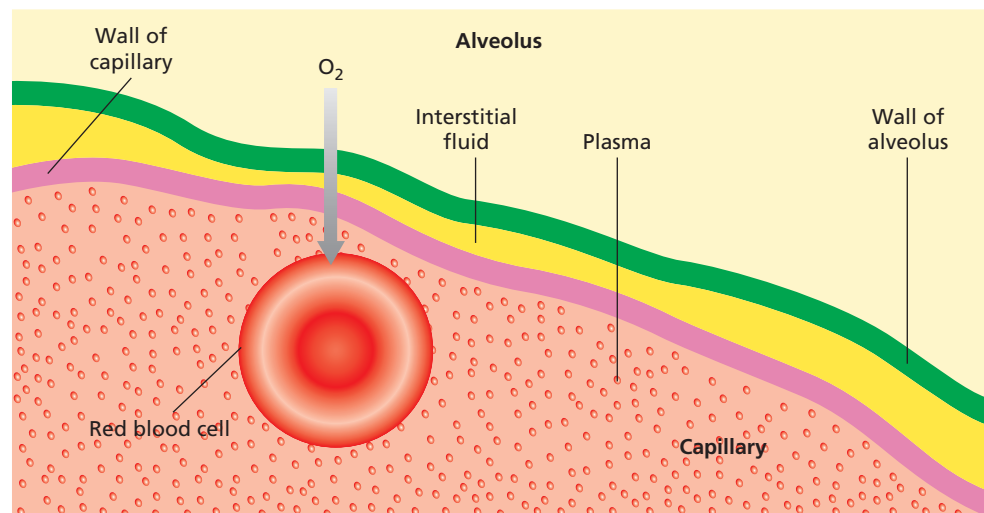


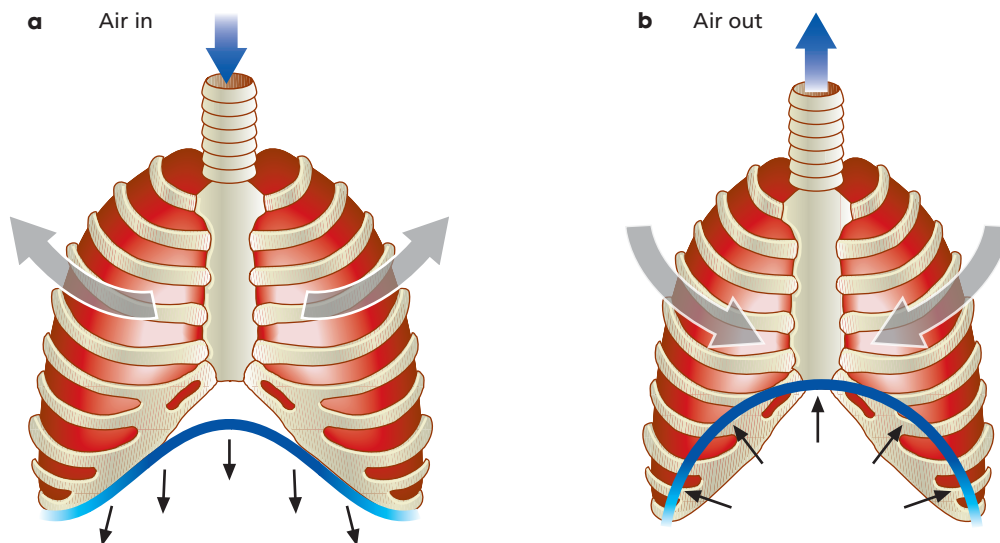
Figure 11.11 ▲
Each alveolar sac is surrounded by a network of capillaries.

Ventilation and breathing rate

In order to extract oxygen from the air by the alveolus, air must enter the gas exchange area of the body and diffuse into cells. Air is moved in and out of the lungs by movements of the ribs and the **diaphragm**. Our rate of breathing is controlled mainly by the level of carbon dioxide in the blood. Most of the carbon dioxide produced by cells in respiration is converted into carbonic acid in the plasma. The bicarbonate ions pass into the blood plasma, where they are transported to the lungs for removal. This lowers the carbon dioxide concentration of the blood.

Figure 11.12 ►
Diffusion distance for oxygen travelling from the alveolus to red blood cells is very short.





◀ **Figure 11.13**
Changes in the size of the chest cavity during breathing. The blue line under the rib cage indicates the diaphragm position when you a) inhale and b) exhale.

Gas exchange in fish

Aquatic animals also need an oxygen supply, but they have the added complication of extracting oxygen dissolved in the water. Fish are able to exchange carbon dioxide and oxygen using **gills**. Gills have the features required of all gas exchange surfaces, as described earlier (see page 275). However, gills do not look like the surfaces associated with air-breathing land animals. The main structure of the gill relies on the buoyancy of the water in which it is immersed to expand its surface area, thus exposing more of the blood supply to the oxygen-drenched water. Gills rely on water flowing over them to ensure maximum oxygen uptake.

There are a variety of ways the gills function in different fish. Most bony fish maintain a flow of water through the gills by what could roughly be described as the equivalent of drinking water but, instead of swallowing the water, they force it out through the gills. Some fish that are very active, such as sharks and tuna, cannot get enough oxygen this way. They swim with their mouths open, letting the water pass in and then flow directly through the gills.

Bony fish have a covering over the gills, the **operculum**, that not only protects the gills but also helps in moving the water into and out of the **opercular cavity**. Cartilaginous fish such as sharks have no operculum, and their six or seven gill slits are very obvious.

Each gill is composed of two piles of leaf-like **filaments**, which project from the gill arch. The upper and lower surfaces of the filaments have numerous gill plates, which greatly increase the surface area of the gill. The gill arch contains an artery that brings deoxygenated blood to the gill and each gill plate is well supplied by capillaries that branch from this artery. As the water flowing over the gill has a higher concentration of oxygen and a lower concentration of carbon dioxide, oxygen diffuses into the blood capillary and carbon dioxide diffuses out. The oxygenated blood is carried away from the gill and the excess carbon dioxide is washed away from the gill by the incoming water.

If fish are taken out of the water, their gills collapse because air does not provide the support that water does. Although air contains more oxygen than water, the fish will die out of the water as the surface area available to exchange gases decreases dramatically and the gills are no longer kept moist.

▼ **Figure 11.14**
In bony fish, gills are located under a bony lid. Neighbouring filaments touch, so that the water flow over them is directed past gas exchange surfaces before being exhaled.



Altitude sickness

Climbing at high altitude (above 2500m) too quickly does not allow the body adequate time to adjust to reduced oxygen levels and air pressure. This results in a condition known as hypobaric hypoxia, which reduces the amount of oxygen reaching the body's tissues. In severe cases, fluid leaks out of the capillaries in the brain or lungs. If it affects the brain, the excess fluid causes the brain to swell in its bony case. Fluid leaking into the lungs increases the diffusion distance for gases, reducing oxygen uptake by the blood. Both conditions may result in death.



Corbis/Robert Boesch

Figure 11.15 ▲

Climbers on the summit of Mt Everest

QUESTION SET 11.3

Remembering

- 1 List the characteristics of an efficient gas exchange surface.
- 2 Describe the function of the surfactant in mammalian lungs.
- 3 Describe how carbon dioxide is carried in the blood.

Understanding

- 4 Explain why living things need to continually exchange gases with their external environment.
- 5 Explain what happens to the ribs and diaphragm during inhalation and exhalation.
- 6 Is the air inside the bronchioles part of the internal or external environment? Explain your answer.
- 7 Explain how the mammalian breathing rate is controlled.

Analysing

- 8 Devise a table to show how each of the characteristics of an efficient gas exchange system is achieved in fish and mammals.

Obtaining nutrients and energy

The type of diet an animal eats determines the features that the animal possesses in order to obtain its food. Vertebrates have **dentition** that suits their diet. For example, a seal that feeds predominantly on fish has sharp, gripping teeth to grab hold of the slippery surface of the fish.

All organisms require access to nutrients that must eventually reach cells.

Importance of digestion

So how exactly does that salad roll you had for lunch or the fish ingested by the seal or the leaves ingested by the koala make their contributions to the cells?

In order to answer this question we will first investigate the nutritional needs of animals, specifically humans. **Digestion** in the human involves the unwrapping and breakdown of large 'packages' of energy and nutrients. The largest packages are in the form of the food taken in, such as bread, meat, butter, fruit and vegetables. The nutrients within these packages, however, are mainly the complex forms of substances such as carbohydrates (starch and cellulose), proteins, and lipids (oils and fats). These complex forms are too large to pass across the cell membrane and into the cells where they are required. Thus, digestion is the process whereby the large complex molecules that make up our food are broken down into simple substances that are small enough to be able to move across the cell membrane and into the internal environment. Digestion takes two forms:

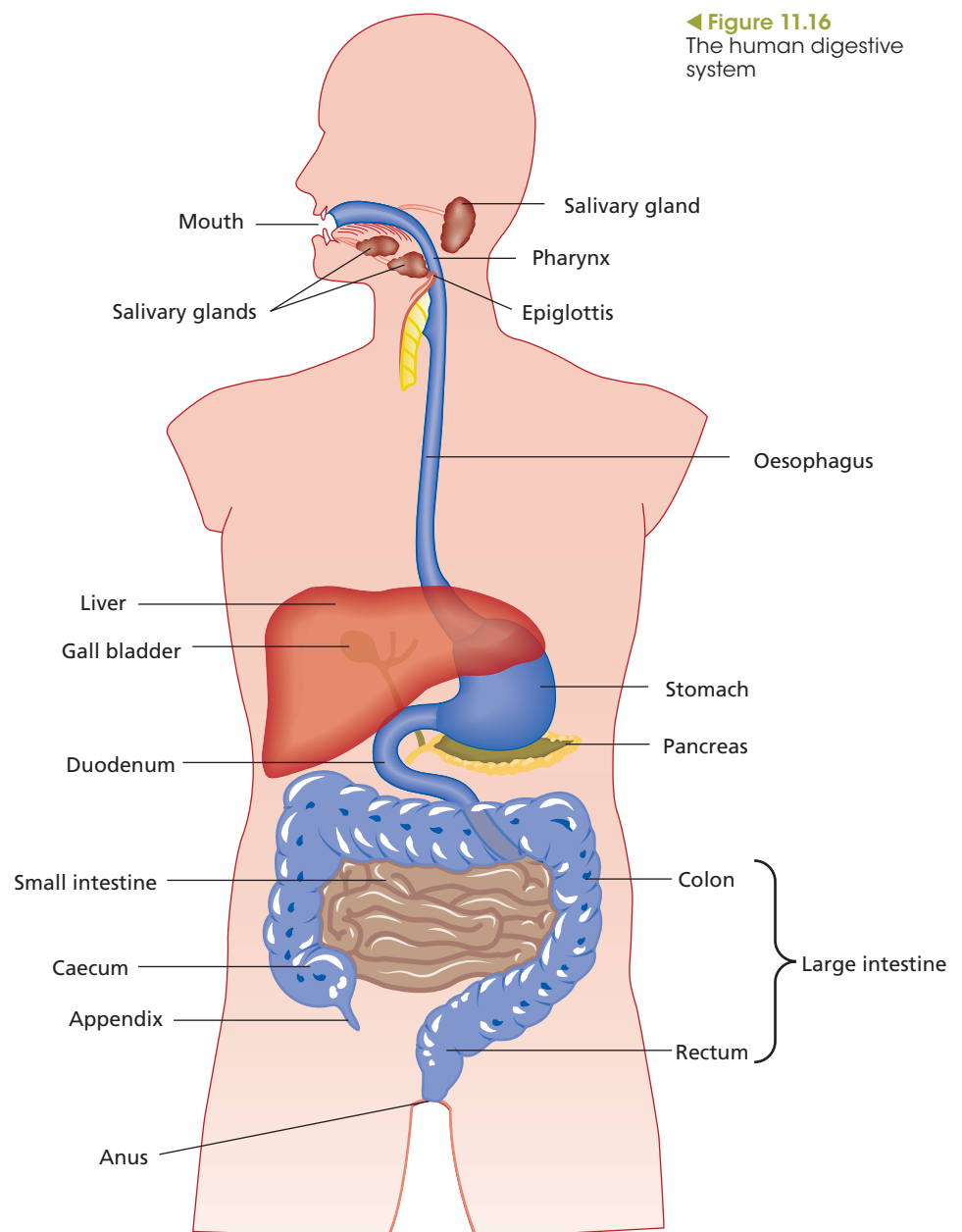
- 1 **Mechanical digestion** (physical digestion) is when large pieces of food are broken down into smaller pieces of food through chewing or muscular movement in the stomach. The aim of this is to increase the surface area of the food so it can be acted on by enzymes in chemical digestion.
- 2 **Chemical digestion** is when enzymes break down complex substances into their simplest forms, such as carbohydrates to glucose, proteins to amino acids, and lipids to glycerol units and fatty acid chains. These are chemically different substances.

Digestion takes place in the **digestive system** (also called the **gastrointestinal tract**). The digestive system has four main roles that involve the functioning of specialised structures.

- 1 **Ingestion**: the taking in of nutrients
- 2 **Digestion**: the breakdown of complex organic molecules into smaller components by mechanical and chemical means
- 3 **Absorption**: the taking up of digested molecules into the internal environment of the cells of the digestive tract
- 4 **Egestion**: the removal of waste food materials from the body.

In humans, the digestive system is one long tube made up of many different types of tissues, and is open at both ends. Humans consume a varied diet and their digestive system deals with this variety in a specific way. The structures involved are similar in all mammals, with some individual adaptations for specialist feeders. We will explore those adaptations later in this chapter.

◀ **Figure 11.16**
The human digestive system



Ingestion

Most animals ingest food through the mouth. Once food has entered the mouth, mechanical and chemical digestion commences – the first stage of unwrapping. The molar teeth at the rear of the mouth help to grind the food into smaller pieces. The **tongue**, a strong muscle, helps to move the food around the mouth, thereby increasing food contact with the teeth. As this is happening, the enzyme **amylase** is secreted from salivary glands situated near the base of the tongue. This enzyme is present in saliva and is specifically suited to the chemical breakdown of complex carbohydrates, such as starch, into simpler carbohydrates, such as maltose. The action of the teeth increases the surface area of the food for greater action by the enzyme.

Figure 11.17 ▼
Various animals and their methods of getting food into their mouths



istockphoto/Marek Dzik



Alamy/Chris A. Cumley



Dreamstime.com/Peter Mikula

Digestion and secretion

Once the food is ready for swallowing, it moves to the back of the mouth and, with the assistance of the tongue, is pushed into the **oesophagus**. The oesophageal opening is extremely close to the opening of the trachea (windpipe). A small flap of tissue, the **epiglottis**, closes off the trachea, preventing food from entering the respiratory tract. The food moves down the oesophagus with the aid of unidirectional muscular contractions known as **peristalsis**. Some chemical digestion of starch will continue here until the food reaches the **stomach**.

In the stomach

The stomach is a muscular part of the gut with the potential to stretch significantly. In animals with specialised diets, this is a very useful ability. The vegetarian koala and wombat eat vast amounts of low-energy food and require a space where the digestion of sclerotic (hard) material can take place. Circular muscles called sphincters regulate the movement of food into and out of the stomach. Sphincters act like the drawstrings on a bag; when the sphincter muscles contract, the opening closes.

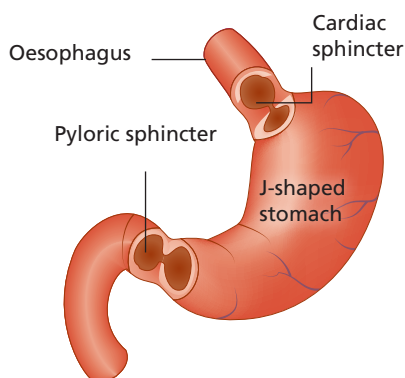


Figure 11.18 ▲
Structure of the human stomach

Mechanical digestion occurs in the stomach as the muscles of the stomach wall relax and contract. The food is still in very large pieces and the aim is to allow the enzymes easy access to most of the surface of the food. The main focus of digestion in the stomach is the chemical digestion of proteins.

The presence of food in the stomach stimulates the secretion of **gastric juice** from cells lining the stomach wall. This substance contains mucus, water, hydrochloric acid and the protein-digesting enzymes pepsin and protease. The presence of hydrochloric acid makes the pH of the stomach around 2.0–3.0; the perfect environment for the action of enzymes. These enzymes break down the long-chain polypeptides of proteins into smaller-chain polypeptides.

The mucus secreted in gastric juice acts to protect the epithelial lining of the stomach from being digested by the enzymes because the cells that line the stomach also contain protein.

Food will remain in the stomach for up to 6 hours until it resembles a 'soupy' substance known as **chyme**. However, further digestion along the length of the gastrointestinal tube is required before the amino acids are small enough to be absorbed into the cells of the body.

In the small intestine

From the stomach, digested food then moves into the small intestine by the actions of peristalsis. A small muscular opening, known as the **pyloric sphincter**, found at the base of the stomach, controls the amount of food leaving the stomach and entering the small intestine. This prevents too much food entering the small intestine too quickly. In adult humans, the small intestine is approximately 7 m long, having a surface area of 4500 m²! There are three defined regions in the small intestine:

- 1 the duodenum
- 2 the jejunum
- 3 the ileum.

Secretions from the pancreas enter the duodenum at the top of the small intestine. These secretions, known as **pancreatic juice**, contain a mixture of amylase, trypsin, lipase and bicarbonate. The chyme entering the duodenum is combined with the hydrochloric acid from the stomach. As previously mentioned, the stomach produces great quantities of mucus to protect its walls. The duodenum does not produce mucus but the acid must still be neutralised. Bicarbonate works to neutralise the acidity of the digested food coming from the stomach. This also stops the action of pepsin and protease.

Bile, a substance produced by the liver, passes down the bile duct and into the duodenum. If there is no food to digest in the small intestine, the bile is stored in the **gall bladder**. Bile is involved in the mechanical breakdown of fats. Bile is not an enzyme but instead has a detergent-like action on the fats. It acts mechanically to emulsify fats down into smaller pieces, thereby increasing the surface area for the action of the enzyme lipase. Lipases are enzymes that are produced by the pancreas and act on the fat droplets. They chemically break down fat into fatty acids and glycerol.

The pancreas also produces the enzyme trypsin. This enzyme acts on the long-chain polypeptides and breaks them down to shorter-chain peptides. A second group of pancreatic enzymes, erepsins, completes the digestion of the short-chain peptides by breaking them down into individual amino acids.

From the duodenum, food enters the jejunum and the ileum where digestion of all food nutrients continues. Digestion of proteins continues and eventually the smaller amino acids are available for absorption into the blood and from there to the body cells. Carbohydrates continue to be broken down into simple sugars such as glucose, which can also be absorbed.

- Mechanical digestion increases the surface area across which digestive enzymes may act.
- Chemical digestion describes the action of the digestive enzymes on the nutrients, ensuring they are at a molecular form and size that is able to be absorbed by cells in the body.

Table 11.3 Summary of chemical digestion

Compound	Enzymes and other chemicals	Product	Location
Carbohydrate	Amylase produced by salivary glands Pancreatic amylase	Maltose	Mouth Duodenum
Maltose	Maltase produced by small intestine	Glucose	Jejunum
Protein	Pepsin produced by stomach Hydrochloric acid produced by stomach to lower pH to optimum level for pepsin Mucus to protect the stomach from hydrochloric acid and pepsin	Peptides	Stomach
Long-chain peptides	Trypsin produced by pancreas to convert long-chain peptides to short-chain peptides Bicarbonate to neutralise hydrochloric acid from stomach	Short-chain peptides	Duodenum
Short-chain peptides	Peptidases produced by the intestinal cells to complete protein digestion	Amino acids	Jejunum
Lipids (fats)	Lipase produced by the pancreas Bile produced by the liver and stored in the gall bladder; acts as detergent to increase the surface area available to the action of lipase	Glycerol and fatty acids	Duodenum

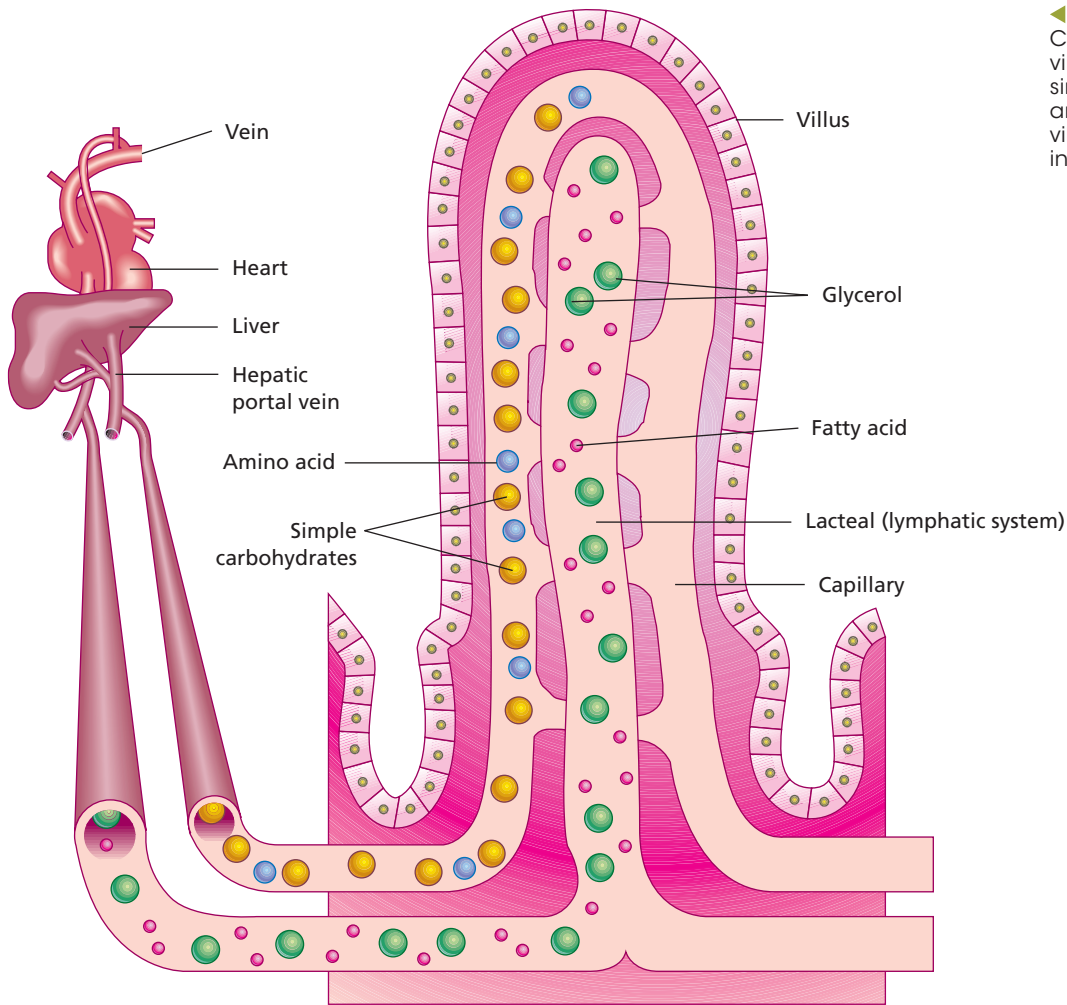
Absorption

The digestive tract discussed so far is all part of the external environment. The products of digestion are now ready to be removed from the digestive tract and transported across a plasma membrane and into the internal environment of the bloodstream. Absorption of nutrients can occur along the length of the gut. Substances such as alcohol and some drugs can be absorbed through the stomach wall into the bloodstream. Because of this, the effects of such substances are felt very quickly. However, most nutrients are absorbed along the length of the small intestine.

In the small intestine, nutrients such as glucose, amino acids, fatty acids and glycerol move from the gut into the blood by means of diffusion and active transport. The structure of the small intestine is perfect for the uptake of nutrients. The lining is moist and thin with a rich supply of blood vessels. Special structures, known as **villi**, project from the surface of the small intestine tissue, increasing the surface area of the gut lining and thus facilitating efficient absorption. Each villus is supplied with a network of capillaries that intertwine with lymph vessels called **lacteals** that transport materials. Glucose and amino acids are absorbed into the capillary network; fatty acids and glycerol are absorbed by the lacteals and enter the lymphatic system. Along with the digested nutrients, water absorption also occurs in the small and large intestines.

Egestion

The **large intestine** is the final length of the gut. This section consists of two main parts, the colon and the rectum. The main functions of the large intestine are to compact undigested food material, such as dietary fibre, and to absorb water and some salts back into the body. Bacteria present in the colon act on the undigested matter, producing vitamins A and K. These vitamins are absorbed through the lining of the large intestine. Peristalsis continues through the large intestine, pushing the waste material into the rectum. This waste material, **faeces**, is eliminated from the body through the anus.



◀ **Figure 11.19**
Cross-section of a villus. Amino acids and simple carbohydrates are absorbed into the villi lining of the small intestine.

Different ways of accessing nutrients

Not many animals have the same **omnivorous** diet that most humans have. Differing diets call for different digestive systems.

Koalas exclusively eat the leaves of a few species of eucalypt trees. Eucalypt leaves are very tough and difficult to digest due to the large amount of cellulose they contain. Cellulose is one of the more complex carbohydrates and thus requires a specialised plan of action so that all possible nutrients are extracted from it. And as we have seen with the digestion of proteins and starch, a specific enzyme is required to break down cellulose; this is known as cellulase. This enzyme is secreted by a variety of micro-organisms, and is not produced by the digestive system of any mammals. As well as being rare, cellulase requires a large surface area on which to work and it takes time to perform. Mechanical digestion increases the surface area exposed to the action of enzymes that carry out the necessary chemical digestion.

The blue whale, despite being the largest mammal alive on the planet, has a diet primarily of krill (*Nyctiphanes australis*). This is a very small crustacean whose main nutritional value is in the protein and lipids it provides. The digestive processes of the whale ensure that the **biomacromolecules** are digested to their component parts of amino acids, and fatty acids and glycerol units, respectively. They are then small enough to enter the cells of the organism as required.



▲ **Figure 11.20**
Koalas chew their food until it is a fine paste in order to provide a large surface area for the action of cellulase.

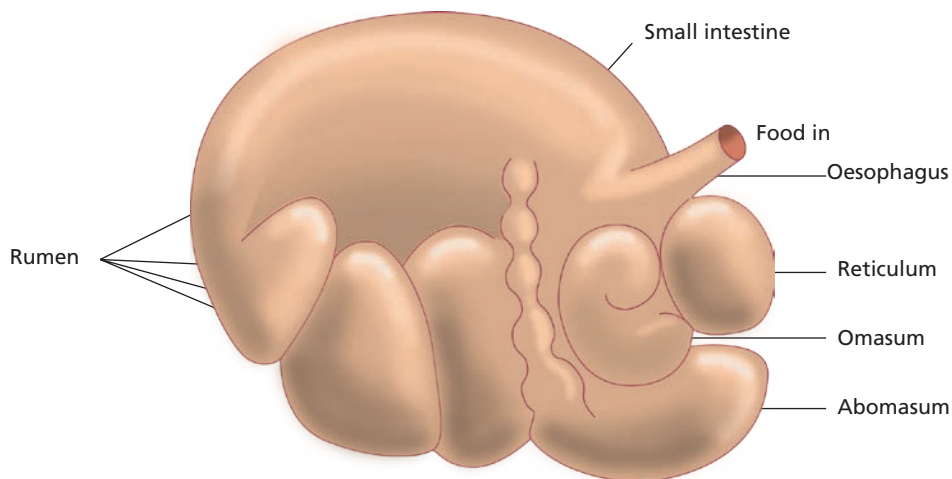


Figure 11.21 ▲
A cow's stomach

Ruminant or cud-chewing mammals such as the cow have an enlarged stomach at the lower end of the oesophagus. This is divided into four chambers, of which the **abomasum** is the true stomach. Grass and other plant material is ground up by the molars and then passed down the oesophagus into the first chamber, the **reticulum**. Here it forms into balls of cud that are regurgitated and chewed again when the cow is not actually feeding. When the cud is swallowed for the second time, it passes into the **rumen**, the

largest chamber. Here it mixes with great numbers of cellulose-digesting bacteria and with large amounts of saliva added from the salivary glands. The rumen is similar to a fermenting chamber and, in anaerobic conditions, the cellulose is chemically digested, releasing carbon dioxide and methane gas as by-products. The contents of the rumen empty into the duodenum, via the **omasum**, where the soluble products of digestion are absorbed.

Non-ruminant herbivorous mammals also possess fermentation chambers where bacteria digest cellulose. In horses and rabbits they are the caecum and appendix, organs situated at the point where the small intestine joins the large intestine. However, there are some disadvantages to having a fermentation chamber near the end of the gut. Food isn't regurgitated and the products of digestion cannot be returned to the small intestine for absorption. To make up for this, rabbits and hares re-ingest their own faeces. If rabbits are deprived of eating their soft faeces, they show signs of nutritional deficiency.

Another group of mammals with specialised dietary requirements are vampire bats. These animals ingest large quantities of mammalian blood very quickly. To accommodate this, they have a very flexible stomach that can expand quickly. As their diet is so specialised, vampire bats do not have an extensive digestive system. The digestion of the protein in the blood occurs in the stomach, as does the absorption of the products.

QUESTION SET 11.4

Remembering

- 1 Describe the main functions of the digestive system. Name the end products.
- 2 List, in order, the structures that food must pass through in a typical vertebrate digestive system.
- 3 State the function of the following enzymes: amylase, pepsin, trypsin and lipase.

Understanding

- 4 Describe the two ways that absorbed nutrients leave the intestine and get into body cells.
- 5 Name structures that assist in increasing the surface area of the digestive system. Explain how their features facilitate absorption.
- 6 Suggest a reason for why cows chew their cud.

Analysing

- 7 Design a summary table for digestion, listing each organ and its function.
- 8 Is the material egested from the digestive system part of the internal or external environment? Give reasons for your answer.
- 9 Compare the structure of the cow's stomach to that of the human stomach. Account for the differences.
- 10 Compare the human digestive system to that of one other organism (not a cow).

The excretory system

Like most machines, the human body is not 100% efficient at converting raw materials into useful substances or forms of energy. As you have learned, cellular respiration produces carbon dioxide that is dissolved in the blood and leaves the body via the respiratory organs. Another source of waste is produced during the oxidation of protein.

Mammals, including humans, must include protein in their diet as a source of amino acids. However, amino acids cannot be stored in the body; those not used are oxidised. The main waste produced by amino acid oxidation is nitrogen, which is excreted as **urea**. Fish excrete their waste nitrogen in ammonia and birds excrete it in uric acid.

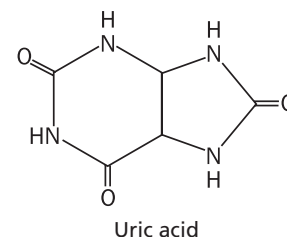
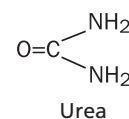
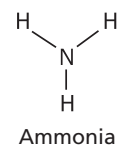
The process that separates the nitrogen-containing amine group from the rest of the amino acid is **deamination**. Part of the amino acid is broken down and used for energy. However, the amine group is converted to ammonia, which is toxic to cells. A build-up of 0.005 mg of ammonia is enough to kill a human. So the human body must either eliminate it very quickly or convert it to a less toxic substance that can be stored temporarily prior to elimination.

In humans, cells in the liver convert the ammonia to less toxic urea. Urea is dissolved in the blood and carried to the organs that excrete it from the body; this is the role of the **kidneys**. In most animals, the urea is diluted by water. In many terrestrial animals, including humans, kidneys need to maintain the delicate balance between the amount of water in the blood and the concentration of solutes dissolved in it.

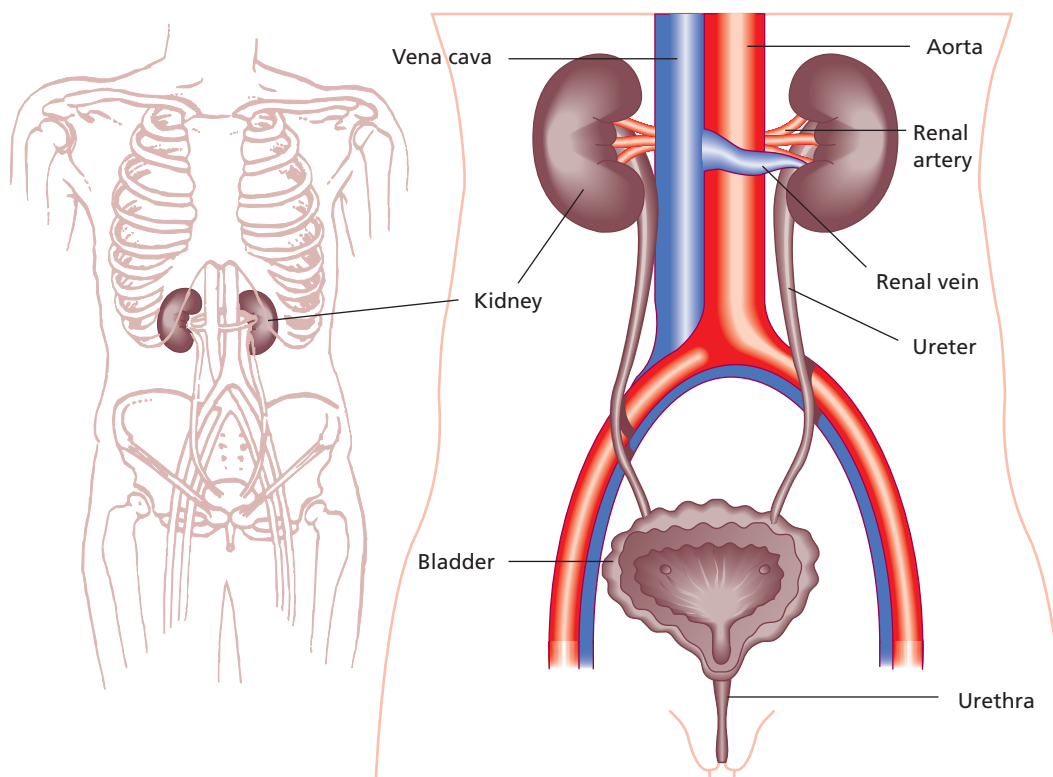
Human kidneys

The main organs of the human excretory system are the kidneys. The kidneys are two bean-shaped organs, each about as big as a fist, located in the back of the upper abdomen at either side of the spinal column (Figure 11.23). A branch of the aorta, the **renal artery**, brings blood containing nitrogenous waste and water, blood proteins, red blood cells and minerals dissolved in the blood plasma to the kidneys. Because the kidneys are the filters that remove wastes from the blood, they may hold as much as 25% of the body's blood at any given time.

The renal artery branches into smaller and smaller vessels until millions of capillaries are formed. Each capillary enters a **nephron** where filtration of the blood occurs.



▲ **Figure 11.22**
Molecules of the main nitrogenous wastes



◀ **Figure 11.23**
The human excretory system

Nephron structure and function

Each kidney consists of approximately one million slender tubules called nephrons, extending from the cortex of the kidney down into the medulla (Figure 11.24). It is at the nephrons that water and solutes are filtered from the blood and amounts to be reclaimed are adjusted. Each nephron consists of a ball of capillaries called a **glomerulus**. The glomerulus is situated inside the **Bowman's capsule**, which looks like a hollow rubber ball that has been pressed in on one side. As the capillaries that form the glomerulus are tightly bound into the smaller Bowman's capsule and pumped by the renal artery, blood in the glomerulus is under very high pressure. This pressure forces out some of the water and all solutes except protein from the blood through tiny pores into the Bowman's capsule. This filtrate has just moved into the external environment as it has passed through the plasma membrane that separates it from the internal environment.

The filtrate moves to the **proximal tubule**, then to a hairpin-shaped **loop of Henle** and finally to the **distal tubule**. A collecting duct, the nephron's last region, is part of a duct system leading to the kidney's central cavity (**renal pelvis**), then into a ureter.

Blood does not give up all of its water and solutes. The unfiltered part flows into capillaries threading around the nephron's tubular parts. In these capillaries, the blood reclaims water and solutes from the tubules. Then it flows into veins and back to the general circulation.

The nephron is the basic functional unit of the kidney.

Figure 11.24 ►
Structure of the kidney
and the functional unit,
the nephron

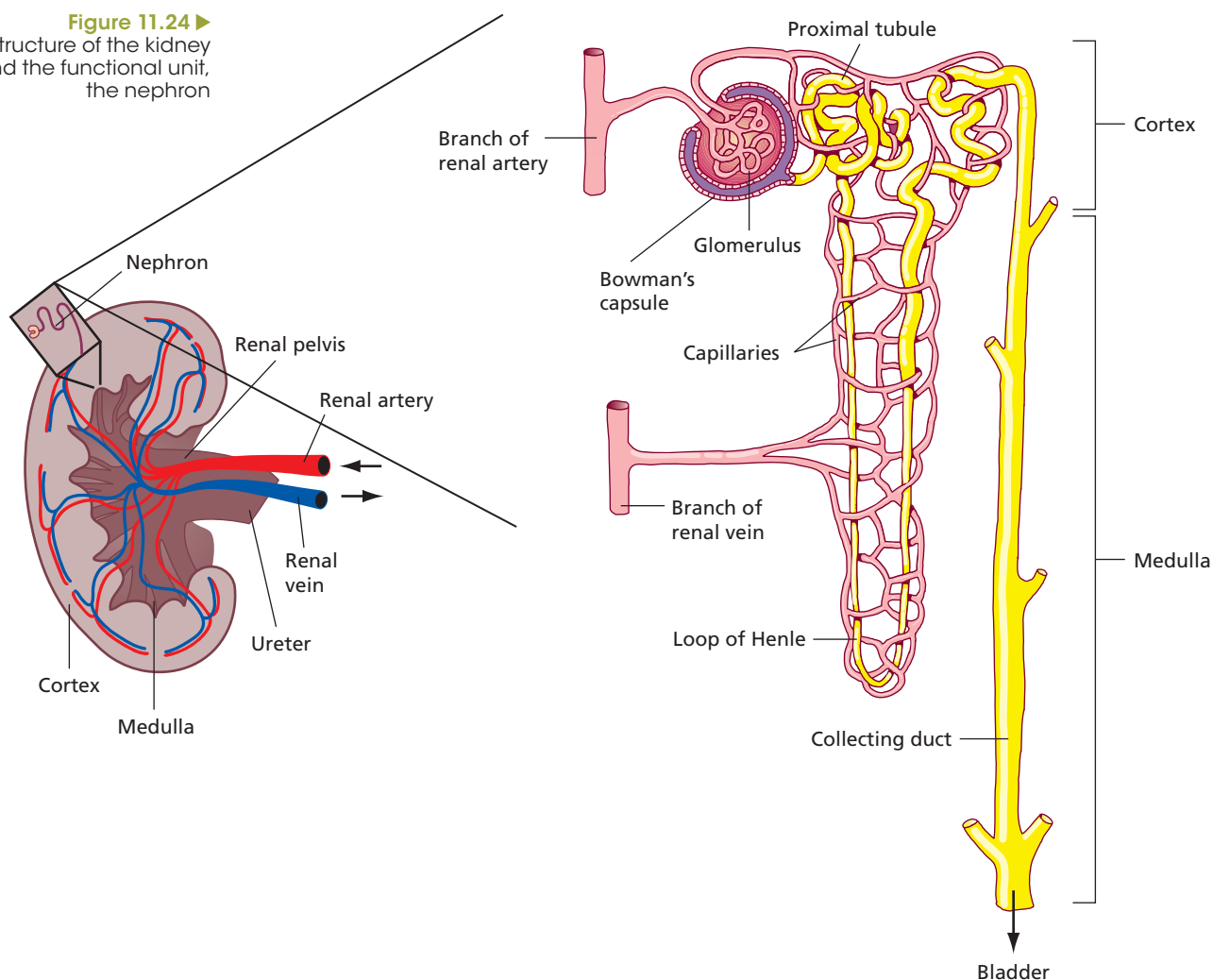


Table 11.4 Summary of the roles of each part of the nephron

Name and structure	Function
Glomerulus: a cluster of capillaries that carry blood from the renal artery	Filtration: pressure of blood forces about one-fifth of its plasma out through the capillary walls, along with everything small enough to pass through: urea, glucose, amino acids and salt, but no red blood cells, platelets or large proteins. These substances enter the Bowman's capsule.
Bowman's capsule: a hollow, cup-shaped end of the nephron with double walls one-cell thick	Filtration: plasma with dissolved materials diffuses into the hollow space of the Bowman's capsule and is passed on to the tubule.
Proximal tubule: a hollow, winding, large-diameter tube	Reabsorption: glucose, sodium ions, calcium ions, phosphate ions and amino acids are actively pumped back into the capillaries. Most of the urea stays inside the tubule.
Loop of Henle (descending): straight part of loop moving away from the proximal tubule	Reabsorption: the concentration of water inside the tubule is greater than that of the surrounding blood vessels; water flows out of the tubule and into the blood by osmosis. Urea stays inside the tubule.
Loop of Henle (ascending): straight part of loop leading to the distal tubule	Reabsorption: sodium is actively pumped out of the tubule and into the blood. Walls of the loop of Henle are impermeable to water, so it remains inside.
Distal tubule: a hollow, winding, large-diameter tube	Reabsorption: a final adjustment of water and dissolved materials. If the body is dehydrated, the permeability of the distal tubule is increased to allow water to return to the capillaries by osmosis. If the body is not dehydrated, permeability decreases, so water remains within the tubule and dilutes the urine that leaves the body.
Collecting ducts: a system of urine-collecting ducts that widen as they near the renal pelvis	Secretion: other wastes are transferred from the capillary network into the duct. Reabsorption: water moves by osmosis from the duct into the surrounding blood.

On average, about 600 mL of fluid flows through each kidney every minute. Approximately 20% of the fluid is filtered into the nephron. If all this fluid were excreted, you would have to drink at least 1 L of fluid every 10 minutes just to maintain water balance. Fortunately, reabsorption ensures that 119 mL of the fluid, along with some essential materials, is put back into the body. The remaining 1 mL becomes urine. Urine is approximately 95% water, with the remaining 5% consisting of dissolved salts, urea and a few other substances. A typical adult excretes approximately 25 g of urea per day. Secretion is the transfer of additional nitrogenous wastes, excess minerals and some other chemicals from the blood into the nephron. Even drugs such as penicillin are secreted.

Kidneys filter water and solutes from blood. Most of this is reabsorbed back into blood, but the excess leaves the kidneys as urine.

The body adjusts for excess water intake by increasing urine output. Conversely, it adjusts for increased exercise or decreased water intake by reducing urine output. The kidneys not only prevent the build-up of wastes but also help maintain water balance by controlling the volume, composition and pressure of body fluids.



EXCRETORY PROCESS

Watch the video and draw up a table that summarises the excretory process in mammals.

Tiny capillaries, large surface area

If all the capillaries in the glomeruli were spread out as a flat sheet, the total surface area would be about 2 m², roughly equal to the surface area of skin in an average adult human. This partly explains how the kidneys are able to filter about 180 L of plasma per day.

EXPERIMENT 11.1

DISSECTION OF A KIDNEY

The excretory organ of mammals is the kidney. Its main function is to filter blood. It also has a role in maintaining specific concentrations of ions and water within the body. These concentrations may vary according to external environmental conditions.

Aim

To become familiar with the structure of the mammalian kidney

Materials

- sheep or pig kidney
- scalpel
- cutting board

Procedure

- 1 Observe the outside appearance of a kidney.
 - a Draw the external shape of the kidney, including any tubes that may be attached to it.
 - b Using Figure 11.24 (structure of a kidney), label any structures you observe.
- 2 Cut through the middle of the kidney lengthwise to show the cortex, medulla and pelvis.
 - a Make a second drawing of the dissected kidney and label the structures that you see.

Results

Include your two drawings of the kidney. Ensure it is clearly labelled including the vessels leading into and out of the structure.

Analysis of results and Discussion

- 1
 - a Name the vessels through which blood enters the kidney.
 - b Name the vessels through which blood leaves the kidney.
 - c Identify and describe what substance surrounds the kidney that makes the identification of the vessels difficult to discern.
- 2 Describe how the pelvis of the kidney differs from the rest of the organ. Name the structure that leads away from the pelvis. Where does it lead to?
- 3 In which structures of the kidney would you expect to find the glomerulus, the loop of Henle and the collecting ducts? Explain your answer using reference material.
- 4 Give reasons why the presence of blood in the urine could indicate a damaged kidney. Explain which structure would most likely be damaged in this instance.



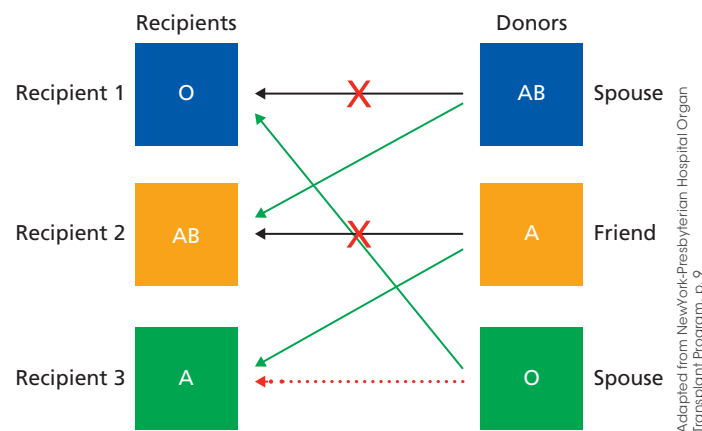
Figure 11.25 ▲
Dissecting a sheep's kidney

Case study

Kidney donation

For many years, licenced drivers have been encouraged to complete documentation that allows their organs to be donated in the unfortunate circumstance of a fatal accident. Where appropriate, the organs are transplanted into recipients who would otherwise die from an impaired or non-functioning organ. Research over the years has indicated, in the area of kidney transplantation specifically, that organ donation from live donors offers prolonged normal organ functioning. It has been found that an organ from a perfectly matched sibling donor can function successfully for up to 35 years. From a less perfectly matched donor, the life span of the organ is 20 years. However, deceased donor transplants see the kidney maintaining function for a maximum of 10 years. Research indicates that the most long-lasting and successful transplants are from living relatives. Occasionally, however, the match between the related donors and recipients are not perfect, and an altruistic unpaired kidney exchange can occur.

In this situation, incompatible donors are matched and six simultaneous surgeries are undertaken. Most donors are able to return to work within a month of surgery, whereas recipients are mostly back within 3 months of the transplant.



▲ Figure 11.26

An example of an altruistic unpaired kidney exchange. Recipient 1 and his spouse were incompatible on blood type. Recipient 2 and her friend were incompatible on tissue type. Recipient 3 and her spouse were compatible. The three pairs participated in an unpaired exchange so that all three recipients could receive kidneys.

However, in a case where a kidney is required urgently, medical specialists are able to provide other alternatives. Such was the case for 35-year-old Amity Bishop whose kidneys had failed. Family members volunteered, however, none were compatible. A transplant from an incompatible donor would lead to rejection of the organ by Ms Bishop's immune system.

A new procedure, overseen by the Director of Renal Medicine at the Alfred Hospital, Rowan Walker, involved reducing the antibodies in Ms Bishop's blood through a series of plasma exchanges. This meant passing the patient's plasma through a machine that removed the antibodies. The lack of antibodies would allow Ms Bishop to receive her mother's donated kidney and prevent its rejection.

Following the transplant, antibody levels rose again, but, for reasons that are not fully understood, the recipient's body accepted the new kidney.

Source: Hagan, K. (2013). 'Kidney transplant a long-shot life saver'. *The Age*, 23 July

Questions

- 1 Identify the type of kidney transplant that maintains function for the longest time period.
- 2 Describe the common complications that arise when kidneys are used from incompatible donors.
- 3 Explain how incompatible kidney transplant rejection has been overcome at the Alfred Hospital.

Other vertebrate kidneys

All vertebrates have kidneys, but they are not all the same as mammalian kidneys. The kidneys of freshwater fish are larger than those of marine fish because they have to put out a larger volume of urine. Some marine fish lack glomeruli because their urine is formed only by secretion of ions, not by filtration.

The loop of Henle occurs only in mammals and some bird kidneys, enabling these animals to produce highly concentrated urine. In fish, the nephrons lie in a straight line. The Australian spinifex hopping mouse has relatively long loops of Henle, producing very concentrated urine and thus reabsorbing enough water so that the animal hardly needs to drink.

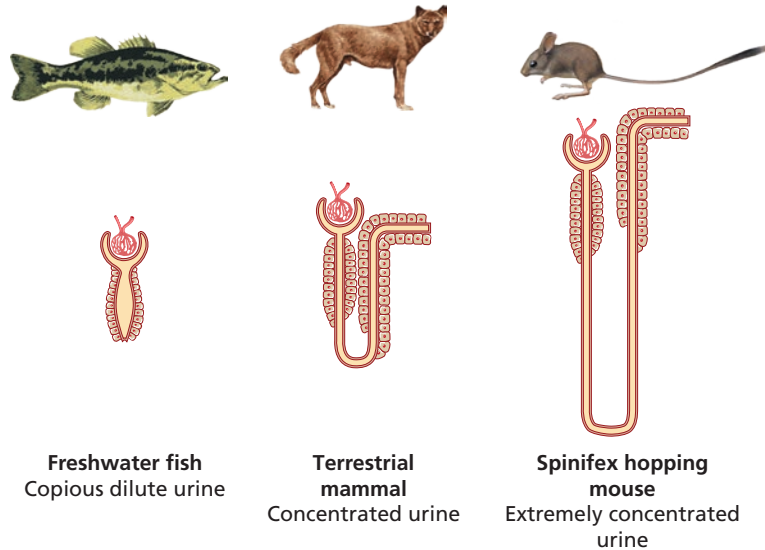
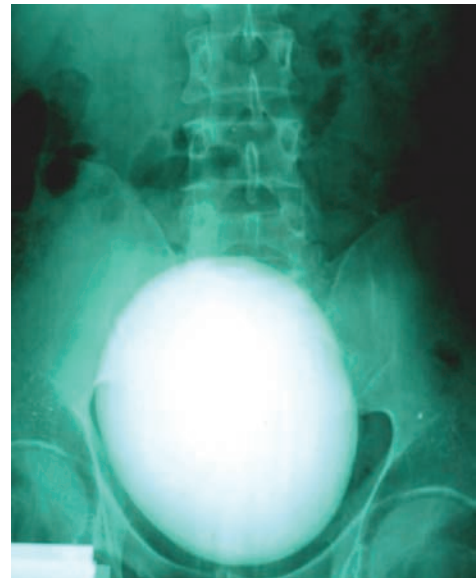


Figure 11.27 ►
A comparison of the structural differences in nephrons between vertebrates

WOW

Kidney stones

Kidney stones form when crystals congregate in the kidney, but pain is usually not experienced until they move into the ureter. The crystals can be composed of many substances, however, they usually form when urine contains too much of a specific substance including uric acid or calcium. Commonly, a lack of urine production or reduced water intake may lead to the formation of the stones. Kidney stones commonly vary in size from a grain of sand to the size of a golf ball. So imagine doctors' surprise to discover one the size of a coconut in a Hungarian man in 2009. When surgeons removed the stone, it measured 17 cm across!



Corbis/TIBOR OLAH/epa

Figure 11.28 ▲
An X-ray photograph showing Sandor Sarkadi's 17 cm kidney stone before its removal

Sweating out waste material

Nitrogenous wastes can also be removed by perspiration. The composition of perspiration or sweat is similar to that of plasma except that sweat does not contain proteins. As the sweat moves along the sweat duct to the skin surface, some sodium and chloride ions are reabsorbed into the bloodstream. Thus, a little urea is lost as sweat.

The salt content of sweat is less than that of blood. Sweat loss, therefore, causes an increase in the salt concentration of the blood. On days of increased sweating, fluid intake must be increased. It is important to remember that sweating is mainly a temperature control mechanism rather than a way of removing nitrogenous wastes.

Scientific literacy: Organ trafficking

International organ trafficking is a growing trade. The increase is down to two factors. First, a reduction in the number of legitimate organs available for transplant – due, in part, to better road safety legislation, which has cut the number of healthy young adults dying prematurely in road traffic accidents. And second, an increase in the number of people waiting for transplants, which have become more routine in recent years. As a result, organised criminals can now make a fortune from unethical clinics who will buy a heart, kidney or pancreas for wealthy patients.

It is now possible to order an organ on the Internet. It's also possible, if you are poor, desperate, and willing to part with, say, a kidney, to broker a deal with traffickers. Recent research by the World Health Organisation (WHO) found that traffickers illegally obtain 7000 kidneys each year around the world.

Organ trafficking operates in various ways. Victims can be kidnapped and forced to give up an organ; some, out of financial desperation, agree to sell an organ; or they are duped into believing they need an operation and the organ is removed without their knowledge. Some victims are murdered to order if a large sum has been paid in advance.

This illegal trade has risen to such a level that an estimated 10 000 black-market operations involving purchased human organs now take place annually – more than one every hour – according to WHO. It estimates that organ trafficking accounts for 5 to 10 per cent of all kidney transplants worldwide.

Bindel, J. (2013) 'Organ trafficking: a deadly trade.' *The Telegraph*, 1 July. © Telegraph Media Group Limited 2013.

Questions

- 1 Suggest reasons why organ trafficking has grown in recent years.
- 2 Discuss the human rights and ethical concerns of organ trafficking.
- 3 Consider the situation of a person giving informed consent to donate their kidney in return for a payment. Create arguments for and against this situation.

QUESTION SET 11.5

Remembering

- 1 List, in order, the pathway that nitrogenous waste takes starting from blood and ending with the collecting duct.
- 2 Name the structure where the final amount of water is reabsorbed.
- 3 Suggest the sorts of substances that are:
a reabsorbed. **b** secreted into the tubule.

Understanding

- 4 Draw a labelled diagram showing the structure of a nephron. On your diagram, indicate where filtration, reabsorption and secretion occur.
- 5 Explain what happens in filtration.

Analysing

- 6 Suggest why the kidney uses more energy proportionally than the heart.
- 7 Explain how the kidneys assist in maintaining water balance in the blood.

CHAPTER SUMMARY

- The circulatory system has four main functions: transporting water, oxygen and carbon dioxide; distribution of nutrients and removal of wastes; maintenance of body temperature; and circulation of hormones.
- The general circulatory system consists of fluid, blood vessels and spaces in which fluid moves, and a pump to push the fluid.
- The cardiovascular system is made up of the heart, blood and blood vessels.
- Blood connective tissue consists of plasma, red and white blood cells, and platelets.
- Red blood cells (erythrocytes) transport oxygen from the lungs to all body regions. They are packed with haemoglobin. They also transport some carbon dioxide from body tissues to the lungs.
- White blood cells (leukocytes) protect against foreign substances that enter the body.
- Platelets are fragments of cells that work to reduce blood loss by forming clots in the case of injury.
- The major organs of the mammalian circulatory system are the heart, arteries, veins and capillaries. Arteries have thick, muscular walls; veins are thinner but have a bigger central cavity and contain valves. Capillaries are very thin and narrow.
- The muscular heart has four chambers: two atria and two ventricles. Structures including valves in the heart ensure blood flows in one direction around the body.
- Lymph vessels carry fluid and plasma proteins from tissues to the heart for return to circulating blood.
- Gas exchange surfaces need to be moist, thin and permeable, have a large surface area and be in a situation where there is a concentration gradient.
- In mammals, air passes into the mouth and nose and travels through to the alveoli tissue in the lungs.
- Breathing ventilates the lungs.
- Gaseous exchange takes place between the alveoli and the capillaries that surround them.
- Fish exchange gases between the external and internal environment via their gills.
- Digestion is the process where large complex molecules are broken down into simple substances. Mechanical digestion is the physical breakdown of substances and chemical digestion is when enzymes, for instance, change the chemical structure of substances to smaller molecules.
- The role of the digestive system includes ingestion, digestion, absorption and egestion.
- Starch digestion starts in the mouth, and protein digestion starts in the stomach. Digestion is completed and most nutrients are absorbed in the small intestine. The pancreas secretes the main digestive enzymes. Bile from the liver assists in fat digestion.
- The excretory system removes nitrogenous wastes from the body. Kidneys are the organs where filtration of blood occurs.
- Kidneys have many nephrons. The start of a nephron is cup-shaped (Bowman's capsule). It continues as three tubular regions (proximal tubule, loop of Henle, and distal tubule that empties into a collecting duct).
- Blood pressure forces water and small solutes from capillaries into the Bowman's capsule. Most of the filtrate gets reabsorbed from the tubules and is returned to the blood.
- Urine forms in the nephron by filtration, reabsorption (a small volume of water and solutes remain in the tubule) and secretion (disposal of urine).

CHAPTER GLOSSARY

abomasum the true stomach of ruminant or cud-chewing mammals, which have a four-chambered stomach

absorption the main role of digestive system; the taking up of digested molecules

albumin plasma molecules

alveoli (alveolus) a tiny air sac located on the end of the bronchioles

amylase a substance present in saliva and secreted from salivary glands near the base of the tongue

aorta the largest artery; takes blood to all parts of the body (i.e. the systemic circulation)

aortic valve the valve between the left ventricle and aorta

arteriole a small artery

artery a blood vessel that takes blood away from the heart

atrioventricular valve the valve between the atrium and ventricle (there are two)

atria two thin-walled chambers of the heart

bile a substance produced by the liver and into the duodenum; it is not an enzyme and is involved in mechanical digestion of fats due to its detergent-like action

biomacromolecule a molecule that has an important structural or functional role in cells

Bowman's capsule the structure in which the glomerulus is found, at the beginning of the nephron

bronchi (singular bronchus) two branches of the trachea that lead to the lungs

bronchiole the smaller tube of the bronchus

capillary a blood vessel found between arteries and veins

carbonic acid most of the carbon dioxide produced in respiration is converted into carbonic acid in the plasma

cardiovascular system see *circulatory system*

chemical digestion a process whereby complex substances are broken down into their simplest forms via enzyme action

chyme 'soupy' contents of the stomach (partially digested food)

circulatory system the heart, blood and blood vessels that ensure nutrients and wastes are carried around the body as needed

closed circulatory system a type of circulatory system; more efficient than an open circulatory system

deamination a process that separates the nitrogen-containing amine group from the rest of the amino acid

dentition the type of teeth of an animal is determined by their diet

diaphragm along with the ribs, an organ that moves air into and out of the lungs

digestion the main role of the digestive system; the breaking down of complex organic molecules mechanically and chemically

digestive system where digestion takes place, also known as the gastrointestinal tract

dissociate the process whereby oxygen separates from haemoglobin at tissue surrounding capillaries

distal tubule the portion of the nephron between the loop of Henle and the collecting duct

egestion the main role of the digestive system; the removal of waste food materials from the body

endothermic describes animals that are able to generate their body heat through biochemical processes within their bodies

epiglottis a small flap of tissue that closes off the trachea, ensuring food travels down the oesophagus to the stomach and not the lungs

erythrocyte a red blood cell that is non-nucleated when mature; contains haemoglobin pigment that transports oxygen from the lungs to the tissues

external environment the environment surrounding a cell outside the plasma membrane

faeces the waste material eliminated from the body through the anus

filament a component of gills; contain numerous gill plates to greatly increase the surface area of the gill

gall bladder the organ that stores bile if there is no food to digest in the small intestine

gastrointestinal tract see *digestive system*

gastric juice a substance produced due to the presence of food in the stomach; contains mucus, water, hydrochloric acid, and pepsin and protease enzymes

gill the gas exchange surface of a fish

glomerulus a network of capillaries located in the Bowman's capsule and are the first step in filtering the blood through the nephron

haemocyanin the respiratory pigment of arachnids and most crustaceans; contains copper instead of iron; blood appears blue when oxygenated

haemoglobin the respiratory pigment of most vertebrates and some invertebrates; contains iron; blood appears red when oxygenated

ingestion the main role of the digestive system; the taking in of nutrients

internal environment all material contained within the plasma membrane

interstitial fluid the fluid between the cells

kidney an organ that excretes the urea dissolved in the blood out of the body

lacteal a vessel that transports materials to and from the villi of the small intestine

large intestine the final length of the gut; consists of the colon and the rectum; functions to compact

undigested food material, and absorb water and some salts back into the body

larynx the voice box

leukocyte a type of white blood cell that forms in both lymph glands and bone marrow, and defends the human body against infectious diseases and foreign particles; the number circulating increases when injury or infection occurs

lipoprotein a biochemical structure that contains both lipids and proteins

loop of Henle the portion of a nephron that connects the proximal convoluted tubule to the distal convoluted tubule

lumen the narrow central cavity of blood vessels

lymph vessel a thin-walled tube with valves that carries lymph around the body

mechanical digestion the process whereby large pieces of food are broken down through chewing or muscular movement in the stomach

nephron the structure of the kidney where filtration of the blood occurs

oedema an excessive build-up of fluid in the body's tissues

oesophagus a vessel that transports food from mouth to stomach

omasum the fourth chamber of the four-chambered stomach of ruminant or cud-chewing mammals

omnivorous describes a diet consisting of both animal and plant foods

open circulatory system the circulatory system in which transport liquid washes freely over the internal organs; less efficient than a closed circulatory system

opercular cavity the cavity in which the gills of bony fish are located

operculum protective covering over the gills in bony fish

oxyhaemoglobin the combination of oxygen and haemoglobin that is carried around the body in red blood cells

pancreatic juice the secretions by the pancreas into the small intestine via the duodenum; contains amylase, trypsin, lipase and bicarbonate

peristalsis the unidirectional muscular contractions that enable food to move down the oesophagus

plasma the pale yellow liquid component of the blood that holds blood cells, proteins and nutrients in suspension

platelet a small cell fragment involved in the clotting property of the blood

proximal tubule the section of the tubule in the nephron that leads from the Bowman's capsule to the loop of Henle

pulmonary circulation the system of blood vessels that leads to and from the lungs

pulmonary valve the valve between the right ventricle and pulmonary artery

pyloric sphincter the small muscular opening that controls the amount of food that can leave the stomach and enter the small intestine

renal artery the branch of the aorta that brings blood (containing nitrogenous waste and water, blood proteins, red blood cells and minerals dissolved in the blood plasma) to the kidneys

renal pelvis the section of the kidney where urine is collected and directed to the ureter

reticulum the first chamber of the four-chambered stomach of ruminant or cud-chewing mammals

rumen the largest chamber of the four-chambered stomach of ruminant or cud-chewing mammals

septum the heart valve that keeps the deoxygenated blood on the right side of the heart from mixing with the oxygenated blood on the left side

stomach a muscular part of the gut that holds food and releases protein-digesting enzymes and hydrochloric acid

surface tension the result of strong cohesive forces between individual water molecules

surfactant the thin layer of fluid that covers the internal membrane of each alveolus

systemic circulation the general body system of blood vessels

tongue an organ that helps to move the food around the mouth and increases food contact with the teeth

trachea an organ that carries gases to the lungs via the bronchi and bronchioles; it is strengthened with rings of cartilage to stop it collapsing

urea the main waste produced by amino acid oxidation is nitrogen, which is excreted as urea

valve a flap that ensures blood is kept flowing in one direction through the heart

vein a blood vessel that carries blood to the heart

vena cava the vein that delivers deoxygenated blood from the general body system to the right atrium

ventricle one of the two muscular chambers of the heart

villi the projections from the surface of the small intestine that increases its surface area and thus increases absorption

CHAPTER REVIEW QUESTIONS

Remembering

- 1 Name the organs that make up the following body systems.
 - a Cardiovascular system
 - b Circulatory system
 - c Respiratory system
 - d Digestive system
 - e Excretory system
- 2 Name where in the body the following processes occur.
 - a Production of urine
 - b Absorption of amino acids
 - c Absorption of water from the products of digestion
 - d Absorption of oxygen into the blood stream
 - e Digestion of protein
 - f Digestion of complex carbohydrates
 - g Ingestion

Understanding

- 3 List the body systems that ensure all cells receive or achieve the following.
 - a Oxygen
 - b Amino acids
 - c Lipids
 - d Removal of waste products from the blood
- 4 The gut is a term commonly used to describe the gastrointestinal tract. Prepare a table that lists all the sections of the gut and describes what occurs in each section.
- 5 Describe the differences in structures involved for systemic and pulmonary circulation.
- 6 Valves are membranous structures found in some blood vessels and between various chambers of the heart. Explain how the material from which they are made helps them undertake their function.
- 7 Visit the weblink, Structure and function of the heart, and complete the quiz.
- 8 Define the following terms and describe their role in the human kidney.
 - a Renal pelvis
 - b Nephron
 - c Loop of Henle
 - d Collecting duct



STRUCTURE AND FUNCTION OF THE HEART

Take this short quiz to test your knowledge of the various heart structures.

Applying

- 9 You have eaten a very large meal of fish. Fish contains a lot of protein and a small amount of fat. Outline how the structure and function of your digestive system will assist the delivery of amino acids to your cells.
- 10 If you suffered a major cut to an artery causing severe blood loss, part of your treatment after stopping the bleeding would be to wrap you in a thermal blanket and place an intravenous drip into a vein. Explain why these two steps are necessary.

Analysing

- 11 Cows do not have canine teeth, but have many flat molars for grinding their food. Explain how this assists them in gaining their nutrition.
- 12 Animals that live in hot, dry environments have nephrons that have a long loop of Henle. Explain how this is an adaptation to their environment.

Evaluating

- 13 The major function of the large intestines is to absorb water. If you are suffering from diarrhoea you would absorb 10 mL of water over 4 hours. If you are suffering from constipation you would absorb 100 mL over 4 hours. Predict the amount of water you would absorb if you were not suffering from either condition.
- 14 When fat droplets meet, they clump together to form a large droplet. This happens in your arteries. Calcium and other minerals form a deposit on top of the fat, forming plaque. Explain why plaque build-up in the arteries could be life threatening.
- 15 Each year, many babies are born with a ventricular septal defect (VSD) where there is a hole in the septum dividing the ventricles. Explain how this could affect the blood being pumped out of the aorta. What do you think ASD could mean?

Creating

- 16 Horses, cats and humans are all mammals. Their body tissues are remarkably similar in make-up, yet their food sources are very different. Horses are herbivores, cats are basically carnivores and most humans are omnivores. Explain how it is that their body tissues are so similar although their food is so different.